

LATENTFLOW: A GENERAL FRAMEWORK FOR CONDITIONING STOCHASTIC PROCESSES

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ABSTRACT

Stochastic-process models are, as a rule, far easier to simulate than to condition. Non-linear observations, non-Gaussian likelihoods, black-box information, and global constraints all induce intractable conditional laws, requiring bespoke, model-specific constructions. We introduce LATENTFLOW, a single framework for conditioning stochastic processes, with no learned neural approximations and no training. Our starting point is to write the stochastic process as the deterministic image of a tractable latent innovation, $f_0 = T_\vartheta(\xi_0)$, with ξ_0 sampled from a simple reference distribution. This reduces process-level conditioning to latent-space inference: pull the likelihood back through T_ϑ , sample the resulting latent law with a tractable guided probability flow, and push the samples forward. This construction is provably exact at the level of the target law; in practice, approximation enters only through finite terminal noising, Monte Carlo guidance, and time discretisation of the continuous-time dynamics, each of which is explicit and systematically reducible. As LATENTFLOW is training-free, conditioning reduces to solving a single reverse-time SDE. This enables conditional sampling in seconds on a single desktop CPU across model classes that have never shared a scalable method: classical spatial priors, nonlinear stochastic dynamics, mechanistic models from the physical and life sciences, stochastic PDEs, heavy-tails and extremes, point and discrete-state processes, and neural or simulator-defined processes.

1 INTRODUCTION

Sampling from a stochastic process only requires running it forward; conditioning it, until now, has remained the preserve of bespoke, model-specific constructions. Examples include diffusion paths conditioned on fixed endpoints or partial observations (Delyon & Hu, 2006; Schauer et al., 2017); functions with shape (Riihimäki & Vehtari, 2010) or physics-informed (Chen et al., 2021; Hamelijnck et al., 2024) constraints; and simulator outputs conditioned on indirect measurements, rare events, or user-defined criteria (Botev & L’Ecuyer, 2020; Finzi et al., 2023). In each case the conditioning information cannot be absorbed into the model in closed form, demanding its own specialised, expensive, or approximate sampler; for instance, those based on MCMC (Andrieu et al., 2003), SMC (Doucet et al., 2001), variational inference (Blei et al., 2017) or Laplace approximations (Rue et al., 2009).

Recently, Moss et al. (2026) introduced FLOWGP, which addresses conditioning for Gaussian process (GP) priors under intractable information by recasting conditional sampling as a differential equation with closed-form Gaussian dynamics and a likelihood-dependent guidance term. The resulting sampler applies to any likelihood that can be evaluated pointwise, and requires no trained network to approximate the guidance field. Underlying the model is a simple mechanism: the GP sample is the linear, deterministic image of a Gaussian innovation, and conditioning is a re-weighting of that innovation’s density by the pulled-back likelihood.

Our key observation is that Gaussianity of the stochastic process prior is not the essential ingredient in this construction; instead, it is the availability of a tractable *latent innovation representation* on which guided dynamics can be defined. Rather than constructing

conditioning dynamics directly in the original process space, this allows us to express the process as the deterministic image of a tractable innovation variable, pull the conditioning information back through this generator, and sample the resulting reweighted law in latent space. The conditioned process is then obtained by pushing the guided samples in the latent space forward through the same generator. We refer to this method as LATENTFLOW. Our construction does not require learning a process prior, score model, or conditional generator (e.g., Song et al., 2021; Chung et al., 2023; Zammit-Mangion et al., 2025); the sampler is built directly from the known map T_θ and pointwise evaluations of the pulled-back likelihood. Our method also provides a natural way to learn the parameters of the process generator.

We obtain formal theoretical guarantees for LATENTFLOW, and demonstrate its application to a wide variety of stochastic process priors. These include GPs; heavy-tailed Student- t and Cauchy-convolution processes; discrete-valued Potts processes; linear and nonlinear diffusion processes including a model for cell-differentiation, a FitzHugh–Nagumo model for excitable neurons, and a Heston stochastic volatility model; mechanistic models of SIR epidemics and Lotka–Volterra predator–prey dynamics; and Allen–Cahn and advection–diffusion SPDEs. See Figure 1 for several examples. Until now, beyond simple or contrived cases, conditioning each of these processes required either low-fidelity approximations or expensive computation. We sample in single-seconds time on a single desktop CPU, providing the first general method for conditioning stochastic processes in real time which is exact up to an explicit and systematically reducible numerical error.

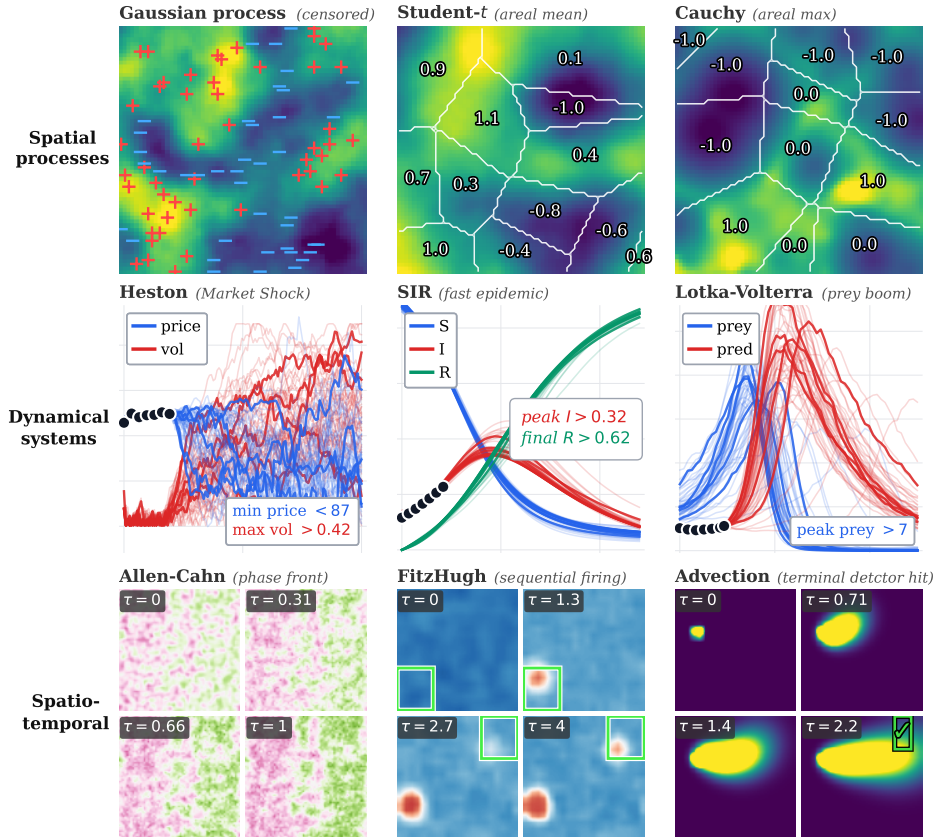


Figure 1: LATENTFLOW across diverse stochastic-process priors and conditions. **Top**: spatial processes conditioned on censored point measurements for a Gaussian process; prescribed areal means for a Student- t process; and prescribed areal maxima for a Cauchy convolution field. **Centre**: SDEs conditioned on observations and events. A Heston model under a joint market-crash and volatility spike; an SIR model under a fast epidemic peak; and a Lotka–Volterra system given a large prey boom. **Bottom**: Spatio-temporal PDEs. Allen–Cahn conditioned on a prescribed phase front (negative left, positive right) at final time; FitzHugh–Nagumo conditioned on sequential source firing (bottom-left then top-right); and an advection–diffusion plume conditioned on reaching a downstream detector (box).

2 CONDITIONING VIA A LATENT REPRESENTATION

Let $\ell \in \mathcal{E}$ denote a stochastic process, viewed as a random element in a measurable space $\mathcal{E}$, and let $f_0 \in E_m$ denote a finite-dimensional discretisation of ℓ . In typical examples, $E_m \subseteq \mathbb{R}^m$ or $E_m \subseteq \mathbb{R}^{md}$, where m is the number of input locations and d is the state dimension at each location. Let $p_\vartheta(f_0)$ denote the prior density of f_0 on E_m , with hyperparameters ϑ . We are interested in sampling the conditional distribution of this process, given some additional information $\mathcal{C}$ encoded by a likelihood $L_\mathcal{C} : E_m \rightarrow [0, \infty)$. The desired conditional density is thus given by

$$\pi_\vartheta(f_0 | \mathcal{C}) = \frac{L_\mathcal{C}(f_0) p_\vartheta(f_0)}{Z_\vartheta}, \quad Z_\vartheta = \int_{E_m} L_\mathcal{C}(f_0) p_\vartheta(f_0) \mathrm{d}f_0. \quad (1)$$

Our key assumption is that p_ϑ can be obtained by transforming a simple reference density.

Assumption 2.1. There exists a latent dimension m_ξ , a reference density r on $\mathbb{R}^{m_\xi}$, and a map $T_\vartheta : \mathbb{R}^{m_\xi} \rightarrow E_m$ such that, if $\xi_0 \sim r$, then $f_0 = T_\vartheta(\xi_0)$ has density p_ϑ .

Assumption 2.1 is very mild; after discretisation, it holds for essentially any process of interest. Consider a scalar target with continuous distribution function F . By the probability integral transform, the variable $F^{-1}(R(\xi_0))$ with $\xi_0 \sim r$ has distribution F , where R is the distribution function of r . The Rosenblatt transform (Rosenblatt, 1952) generalises this to multivariate f_0 and writes *any* absolutely continuous law on E_m as the deterministic image of some $\xi_0 \sim r$. The choice of r is arbitrary; we fix $r(\xi_0) = \mathcal{N}(\xi_0; 0, I_{m_\xi})$ throughout. It is worth noting that we use densities here only to simplify the presentation. In fact, we require only a measure-theoretic version of Assumption 2.1, which also applies when the prior is singular or discrete; see Remark B.1 for details.

Once such a representation is available, conditioning the process on $\mathcal{C}$ can be transferred to the latent space via the *latent likelihood* $\Lambda_{\vartheta, \mathcal{C}}(\xi_0) = (L_\mathcal{C} \circ T_\vartheta)(\xi_0)$, i.e. the standard likelihood pulled back to $\mathbb{R}^{m_\xi}$. The corresponding latent target density is

$$\rho_\vartheta(\xi_0 | \mathcal{C}) = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi_0) r(\xi_0)}{Z_\vartheta}, \quad Z_\vartheta = \int_{\mathbb{R}^{m_\xi}} \Lambda_{\vartheta, \mathcal{C}}(\xi_0) r(\xi_0) \mathrm{d}\xi_0, \quad (2)$$

where the posterior is recovered by pushing ρ_ϑ through T_ϑ . That is, if $\xi_0 \sim \rho_\vartheta(\cdot | \mathcal{C})$, then $f_0 = T_\vartheta(\xi_0)$ has density $\pi_\vartheta(\cdot | \mathcal{C})$. We formalise this equivalence in Proposition 2.2, which states that the latent and process targets agree in distribution.

Proposition 2.2. Suppose that Assumption 2.1 holds and that $0 < Z_\vartheta < \infty$. Then, if $\xi_0 \sim \rho_\vartheta(\cdot | \mathcal{C})$, the random variable $f_0 = T_\vartheta(\xi_0)$ has the conditioned process density $\pi_\vartheta(\cdot | \mathcal{C})$.

2.1 EXAMPLES OF LATENT INNOVATION REPRESENTATIONS

Below we provide three examples of stochastic processes that satisfy Assumption 2.1. Further examples, including heavy-tailed and max-stable processes, state-space models and SPDEs, and neural processes and black-box simulators, are given in Appendix A.

Gaussian process. Let $f \sim \text{GP}(\mu_\vartheta, k_\vartheta)$, and let $f_0 = (f(s_1), \dots, f(s_m)) \in \mathbb{R}^m$ denote its values at inputs $s_{1:m}$. Thus $f_0 \sim \mathcal{N}(m_\vartheta, K_\vartheta)$, where $m_\vartheta = (\mu_\vartheta(s_1), \dots, \mu_\vartheta(s_m))$ and $(K_\vartheta)_{ij} = k_\vartheta(s_i, s_j)$. Let L_ϑ be a matrix satisfying $L_\vartheta L_\vartheta^\top = K_\vartheta$. We can then write

$$f_0 = T_\vartheta(\xi_0) = m_\vartheta + L_\vartheta \xi_0, \quad \xi_0 \sim \mathcal{N}(0, I_m).$$

Thus, the latent innovation is the whitened Gaussian vector ξ_0 , and the generator T_ϑ is the usual affine GP whitening map.

Student- t process. Let $f \sim \text{TP}_\nu(\mu_\vartheta, k_\vartheta)$, where k_ϑ denotes the Student- t scale kernel, and let $f_0 = (f(s_1), \dots, f(s_m)) \in \mathbb{R}^m$ denote its values at inputs $s_{1:m}$. We then have

$$f_0 = m_\vartheta + \omega^{-1/2} L_\vartheta v,$$

where $v \sim \mathcal{N}(0, I_m)$ and $\omega \sim \text{Gamma}(\nu/2, \nu/2)$ independently. We can also express f_0 in terms of a standard Gaussian reference. Let $a \sim \mathcal{N}(0, 1)$, and define $G_\nu(a) = F_{\Gamma, \nu}^{-1}(\Phi_N(a))$, where $F_{\Gamma, \nu}$ is the $\text{Gamma}(\nu/2, \nu/2)$ CDF, and Φ_N is the standard normal CDF. Then

$$f_0 = T_\vartheta(\xi_0) = m_\vartheta + G_\nu(a)^{-1/2} L_\vartheta v, \quad \xi_0 = (v, a) \sim \mathcal{N}(0, I_{m+1}).$$

Thus, m Gaussian variables generate the spatial Gaussian component, and one additional Gaussian variable generates the random scale.

SDE-driven process. Consider a process defined via the stochastic differential equation

$$dx_\tau = b_\vartheta(\tau, x_\tau) d\tau + \Sigma_\vartheta(\tau, x_\tau) dw_\tau,$$

where $w = (w_\tau)_{\tau \geq 0}$ is a standard Brownian motion. Define a time grid $0 = \tau_0 < \tau_1 < \dots < \tau_N = T$, with $\Delta_n = \tau_{n+1} - \tau_n$. The Euler–Maruyama discretisation of this SDE is

$$x_{n+1} = x_n + b_\vartheta(\tau_n, x_n)\Delta_n + \Sigma_\vartheta(\tau_n, x_n)\sqrt{\Delta_n}\eta_n,$$

where $\eta_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I)$. If x_0 is random, we also write $x_0 = \chi_\vartheta(\eta_{-1})$, with $\eta_{-1} \sim \mathcal{N}(0, I)$, and omit η_{-1} when x_0 is fixed. The latent innovation is the collection of Gaussian variables $\xi_0 = (\eta_{-1}, \eta_0, \dots, \eta_{N-1})$, with the obvious modification for deterministic x_0 . Meanwhile, the generator T_ϑ is the numerical solver itself: it takes ξ_0 , applies the Euler–Maruyama recursion, and returns the discretised path $(x_0, x_1, \dots, x_N) = T_\vartheta(\xi_0)$.

3 LATENTFLOW: INFERENCE WITH GUIDED LATENT FLOWS

We now develop the method on which the paper rests. Whenever the latent likelihood is differentiable, or its score admits a tractable estimator, a guided diffusion draws samples $\xi_0 \sim \rho_\vartheta(\cdot | \mathcal{C})$. We call the resulting framework LATENTFLOW. The statements provided here are formal; see [Appendix C](#) for a more rigorous treatment.

3.1 NOISING BRIDGE, POTENTIAL, AND GUIDANCE FIELD

We first define a Gaussian noising process on the latent space. Let $\xi_0 \sim r$ and $\varepsilon \sim \mathcal{N}(0, I_{m_\xi})$ be independent of ξ_0 . The variance-preserving (VP) latent noising bridge is given by

$$\xi_t = \alpha(t)\xi_0 + \sigma(t)\varepsilon, \quad (3)$$

where $\alpha(t) = \exp\{-\frac{1}{2} \int_0^t \beta(s) ds\}$ and $\sigma(t)^2 = 1 - \alpha(t)^2$, for a locally integrable function $\beta : [0, 1) \rightarrow (0, \infty)$ such that $\lim_{t \uparrow 1} \alpha(t) = 0$. Thus ξ_0 is the clean latent variable and ξ_t for $t \uparrow 1$ is increasingly noised.

Since $r(\xi_0) = \mathcal{N}(\xi_0; 0, I_{m_\xi})$, we have $\xi_t \sim r$ for every $t \in [0, 1)$. Moreover, the conditional density of $\xi_0 | \xi_t$ is given by

$$r_{0|t}(\xi_0) := p(\xi_0 | \xi_t = u) = \mathcal{N}(\xi_0; \alpha(t)u, \sigma(t)^2 I_{m_\xi}). \quad (4)$$

The additional condition enters through the latent likelihood Λ . For $t \in [0, 1)$, we define the *latent conditioning potential* as

$$\psi(t, u) = \mathbb{E}_{\xi_0 \sim r} [\Lambda_{\vartheta, \mathcal{C}}(\xi_0) | \xi_t = u] = \mathbb{E}_{\varepsilon \sim r} [\Lambda_{\vartheta, \mathcal{C}}(\alpha(t)u + \sigma(t)\varepsilon)].$$

This can be viewed as a smoothed version of the latent likelihood. This function also has a density-ratio interpretation. Let $\xi_t \sim \rho_t$ be the random variable obtained by drawing $\xi_0 \sim \rho_0$ and then applying the VP noising bridge in (3). Then $\psi(t, u) = Z_\vartheta \frac{\rho_t(u)}{r(u)}$.

Finally, suppose $\psi(t, u) > 0$ and that $\psi(t, \cdot)$ is differentiable at u . We define for $t \in [0, 1)$ the *latent guidance field*

$$g(t, u) = \nabla_u \log \psi(t, u) = \nabla_u \log \frac{\rho_t(u)}{r(u)}.$$

This will later be used to steer the dynamics in the direction of the increased expected future likelihood.

3.2 EXACT LATENT DYNAMICS

Our goal is a sampling procedure for $\xi_0 \sim \rho_0$ with $\rho_0 \equiv \rho_\vartheta$. First recall that the family of guided densities $(\rho_t)_{0 \leq t < 1}$ are precisely the one-time marginals of the VP-SDE

$$d\xi_t = -\frac{1}{2}\beta(t)\xi_t dt + \sqrt{\beta(t)} dw_t, \quad \xi_0 \sim \rho_0, \quad t \in [0, 1),$$

where w is a standard $\mathbb{R}^{m_\xi}$ -valued Brownian motion (e.g., Song et al., 2021). By Anderson (1982), the time reversal of these dynamics is also an SDE, given by

$$d\xi_t = \beta(t) \left[\frac{1}{2} \xi_t - g(t, \xi_t) \right] dt + \sqrt{\beta(t)} d\bar{w}_t, \quad (5)$$

where $\bar{w}$ is a reverse-time Brownian motion, and this equation is integrated backwards in t . In particular, if (5) is initialised at $\xi_{t_1} \sim \rho_{t_1}$ and integrated backwards from t_1 to 0, then $\xi_t \sim \rho_t$ for every $t \in [0, t_1]$, and hence $\xi_0 \sim \rho_0$. Since $\rho_{t_1} \Rightarrow \mathcal{N}(0, I_{m_\xi})$ as $t_1 \uparrow 1$, initialising at $\mathcal{N}(0, I_{m_\xi})$ in place of ρ_{t_1} is asymptotically exact, yielding a practical sampler.

The same family of marginal densities can also be generated by a deterministic process known as the probability-flow ODE (see Song et al., 2021). In practice, however, the SDE is often a preferable generating mechanism; see Appendix C for further details.

3.3 MONTE CARLO GUIDANCE

To generate ξ_0 using the reverse-time SDE in (5) we first need $g(t, u) = \nabla_u \log \psi(t, u)$. Typically, neither $g(t, u)$ nor $\psi(t, u)$ are available in closed form. We can, however, obtain estimates of both via a Monte Carlo estimator of (4). For a given pair (t, u) , draw

$$\varepsilon^{(s)} \sim \mathcal{N}(0, I_{m_\xi}), \quad \xi_{0|t}^{(s)} = \alpha(t)u + \sigma(t)\varepsilon^{(s)}, \quad s = 1, \dots, S.$$

Then $\hat{\psi}_S(t, u) = \frac{1}{S} \sum_{s=1}^S \Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(s)})$ is an unbiased estimator of $\psi(t, u)$. If $\Lambda_{\vartheta, \mathcal{C}}$ is differentiable, then

$$\nabla_u \log \psi(t, u) = \alpha(t) \frac{\mathbb{E}_{\varepsilon \sim r} [\nabla_\xi \Lambda_{\vartheta, \mathcal{C}}(\alpha(t)u + \sigma(t)\varepsilon)]}{\mathbb{E}_{\varepsilon \sim r} [\Lambda_{\vartheta, \mathcal{C}}(\alpha(t)u + \sigma(t)\varepsilon)]}.$$

When $\Lambda_{\vartheta, \mathcal{C}} > 0$, this motivates the self-normalised estimator

$$\hat{g}_S(t, u) = \alpha(t) \sum_{s=1}^S \bar{w}_s \nabla_\xi \log \Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(s)}), \quad \bar{w}_s = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(s)})}{\sum_{\ell=1}^S \Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(\ell)})}. \quad (6)$$

Since $\Lambda_{\vartheta, \mathcal{C}} = L_{\mathcal{C}} \circ T_{\vartheta}$, the latent score can be computed by differentiating through the process generator; in practice, this is typically available via automatic differentiation. In particular, when T_{ϑ} and $L_{\mathcal{C}}$ are differentiable and $L_{\mathcal{C}} > 0$, the chain rule gives

$$\nabla_\xi \log \Lambda_{\vartheta, \mathcal{C}}(\xi) = [D_\xi T_{\vartheta}(\xi)]^\top \nabla_{f_0} \log L_{\mathcal{C}}(f_0) \Big|_{f_0=T_{\vartheta}(\xi)},$$

where $D_\xi T_{\vartheta}(\xi)$ is the Jacobian of $T_{\vartheta}(\xi)$. For hard constraints, discontinuous likelihoods, or non-differentiable simulators, this gradient-based form must be replaced by a smooth relaxation, a differentiable surrogate, or a derivative-free guidance estimator; see Section D.7.

In practice, we replace $g(t, u)$ by $\hat{g}_S(t, u)$ and solve (5) numerically. The resulting error decomposes into three terms: terminal initialisation, guidance approximation, and numerical discretisation; see Appendix D. Under a common exact terminal coupling, if the Monte Carlo guidance field is uniformly root- S consistent and the SDE solver has strong order q_{sde} , then the coupled latent grid error is $\mathcal{O}_{\mathbb{P}}(S^{-1/2} + h^{q_{\text{sde}}})$. The error contribution of each can be made arbitrarily small, so the sampling error is limited only by available compute. These bounds pass to process space whenever T_{ϑ} is Lipschitz.

4 GUIDANCE-AWARE PARAMETER ESTIMATION

We have, thus far, treated the parameter ϑ as fixed. Adapting it based on the information $\mathcal{C}$ offers further advantages of our training-free approach. A learned conditional generator or amortised sampler must be retrained as ϑ is updated, or trained in advance over a prescribed range of parameter values. In LATENTFLOW, T_{ϑ} is a pre-defined generative prior, and ϑ enters through the pulled-back likelihood. Parameter estimation can thus be performed using the same latent conditional samples, without retraining a conditional generator.

The normalising constant Z_{ϑ} in (1) and (2) can be interpreted as the evidence for $\mathcal{C}$ under the prior. This suggests the empirical Bayes objective $\mathcal{J}(\vartheta) = \log Z_{\vartheta}$. This balances satisfaction

of $\mathcal{C}$ with the relative-entropy cost of deforming p_ϑ into $\pi_\vartheta(\cdot \mid \mathcal{C})$; see [Proposition E.1](#). This relative entropy cost also admits a dynamic interpretation as the guidance energy; see [Theorem E.5](#). Thus, parameters are compatible with the condition when they can be realised with high likelihood and low guidance energy.

Given $\nabla_\vartheta \mathcal{J}(\vartheta) = \nabla_\vartheta \log Z_\vartheta$, or an approximation thereof, we can update parameters using an outer-loop scheme. Fix ϑ_0 . For each $k \geq 0$, run LATENTFLOW with ϑ_k fixed, and update

$$\vartheta_{k+1} = \vartheta_k + \eta_k \nabla_\vartheta \mathcal{J}(\vartheta) \big|_{\vartheta=\vartheta_k},$$

where $\eta_k > 0$ is the step size. When $\nabla_\vartheta \mathcal{J}(\vartheta)$ is unknown, we require an approximation. If $\Lambda_{\vartheta, \mathcal{C}}$ is differentiable in ϑ and differentiation and integration are interchangeable, then

$$\nabla_\vartheta \mathcal{J}(\vartheta) = \mathbb{E}_{\xi_0 \sim \rho_\vartheta(\cdot \mid \mathcal{C})} [\nabla_\vartheta \log \Lambda_{\vartheta, \mathcal{C}}(\xi_0)].$$

Thus samples from the latent target provide a direct estimator of the conditional-evidence gradient. If the likelihood $L_{\mathcal{C}}$ is not a function of ϑ , then

$$\nabla_\vartheta \log \Lambda_{\vartheta, \mathcal{C}}(\xi_0) = [D_\vartheta T_\vartheta(\xi_0)]^\top \nabla_{f_0} \log L_{\mathcal{C}}(f_0) \big|_{f_0=T_\vartheta(\xi_0)},$$

where $D_\vartheta T_\vartheta(\xi_0)$ is the Jacobian of $T_\vartheta(\xi_0)$. The same gradient can also be represented at intermediate noising times. In particular, we have that

$$\nabla_\vartheta \mathcal{J}(\vartheta) = \nabla_\vartheta \log Z_\vartheta = \mathbb{E}_{u \sim \rho_{\vartheta, t}(\cdot \mid \mathcal{C})} [\nabla_\vartheta \log \psi_\vartheta(t, u)].$$

This intermediate-time gradient can be estimated using the same bridge samples used for guidance; see [Section 3.3](#). In particular, differentiating the estimator of the conditioning potential with respect to ϑ yields

$$\hat{a}_{\vartheta, S}(t, u) := \nabla_\vartheta \log \hat{\psi}_{\vartheta, S}(t, u) = \sum_{r=1}^S \bar{w}_r \nabla_\vartheta \log \Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(r)}), \quad \bar{w}_r = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(r)})}{\sum_{\ell=1}^S \Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(\ell)})}. \quad (7)$$

Thus, the same weighted bridge candidates provide two derivatives: [\(6\)](#) guides the latent state, while [\(7\)](#) adapts the process parameter. Given M particles and selected noising times t_j , a practical estimator is

$$\nabla_\vartheta \mathcal{J}(\vartheta) \approx \frac{1}{MJ} \sum_{i=1}^M \sum_{j=1}^J \hat{a}_{\vartheta, S}(t_j, u_j^{(i)}).$$

5 RELATED WORK

The closest antecedent to our work is FLOWGP ([Moss et al., 2026](#)). Our framework contains FLOWGP as a special case, but also applies to a substantially broader class of stochastic processes; see [Appendix A](#). More broadly, our approach relates to diffusion models, flow matching, and stochastic interpolants, which generate samples via continuous-time transport ([Sohl-Dickstein et al., 2015](#); [Ho et al., 2020](#); [Song et al., 2021](#); [Lipman et al., 2023](#); [Albergo et al., 2025](#)). Unlike pretrained diffusion-model guidance (e.g., [Ho & Salimans, 2022](#)), however, LATENTFLOW does not need to be pre-trained on a neural estimate of an unconditional data score or reverse process. The noising bridge is imposed in innovation space, and the guidance is the gradient of a smoothed pulled-back likelihood under the known latent bridge.

A closely related line of work conditions generative models by performing inference in the latent noise space ([Whang et al., 2021](#); [Holden et al., 2022](#); [Patel et al., 2022](#); [Achituve et al., 2025](#); [Askari et al., 2025](#); [Purohit et al., 2025](#); [Venkatraman et al., 2025](#); [Xia et al., 2026](#)). These methods share the idea of conditioning a generator by changing the distribution of its input noise, but focus only on learned neural generators. Our framework is significantly more general: it includes pre-trained neural generators as a special case, but also extends to a much broader class of genuine stochastic-process priors. The papers above also rely on different, often more expensive, samplers, for instance, Langevin dynamics ([Purohit et al., 2025](#)), HMC ([Patel et al., 2022](#); [Xia et al., 2026](#)), pCN ([Holden et al., 2022](#)), or SMC ([Achituve et al., 2025](#)). LATENTFLOW instead targets the latent posterior using a guided, training-free diffusion process, whose guidance field is constructed from the pulled-back likelihood.

Our work is also related to many other areas, including diffusion bridges, rare-event simulation, and transport-assisted MCMC; see [Appendix F](#) for a more detailed discussion.

6 NUMERICAL EXPERIMENTS

We now present a range of numerical experiments designed to assess the flexibility, accuracy, and scalability of LATENTFLOW. We first evaluate LATENTFLOW on three standard sampling benchmarks; see [Appendix G.1](#) for full details. Across all three benchmarks, we achieve MMD² scores competitive with or better than NUTS. As LATENTFLOW generates independent samples, it scales to problems where standard MCMC either collapses modally or explores very slowly. In this section, we consider higher-dimensional and more challenging processes. All timings are reported on a 13th Gen Intel i9-13900K CPU. Additional results are provided in [Appendix G](#).

6.1 SPATIAL PROCESSES

Spatial processes are central to many applications, including environmental modelling, climate science, and ecology (e.g., [Cressie, 1993](#)). While GPs provide a convenient prior in such settings, many applications exhibit extremal or heavy tails, discontinuities, sharp local anomalies, or discrete observations. This motivates alternatives such as Student- t processes, Lévy random fields, and Potts models ([Potts, 1952](#); [Besag, 1974](#); [Rajput & Rosiński, 1989](#); [Wolpert et al., 2011](#); [Shah et al., 2014](#)). The difficulty here has two sources. The prior alone can break conjugacy as non-Gaussian priors induce conditional laws involving scale mixtures or jump-like innovations ([Walchessen et al., 2025](#)). Additionally, non-linear conditioning statements such as areal maxima, threshold exceedances, or censored measurements do not admit closed-form conditionals, even under a Gaussian prior.

In [Figure 2](#), we use LATENTFLOW to condition four spatial priors on four conditions. The conditioning statements include point observations, regional summaries, and censored observations. The resulting samples qualitatively respect the imposed conditioning information, whilst also preserving the distinctive behaviour of each prior. The speed of LATENTFLOW is notable here: conditional samples are generated in just 3 – 4 seconds per sample.

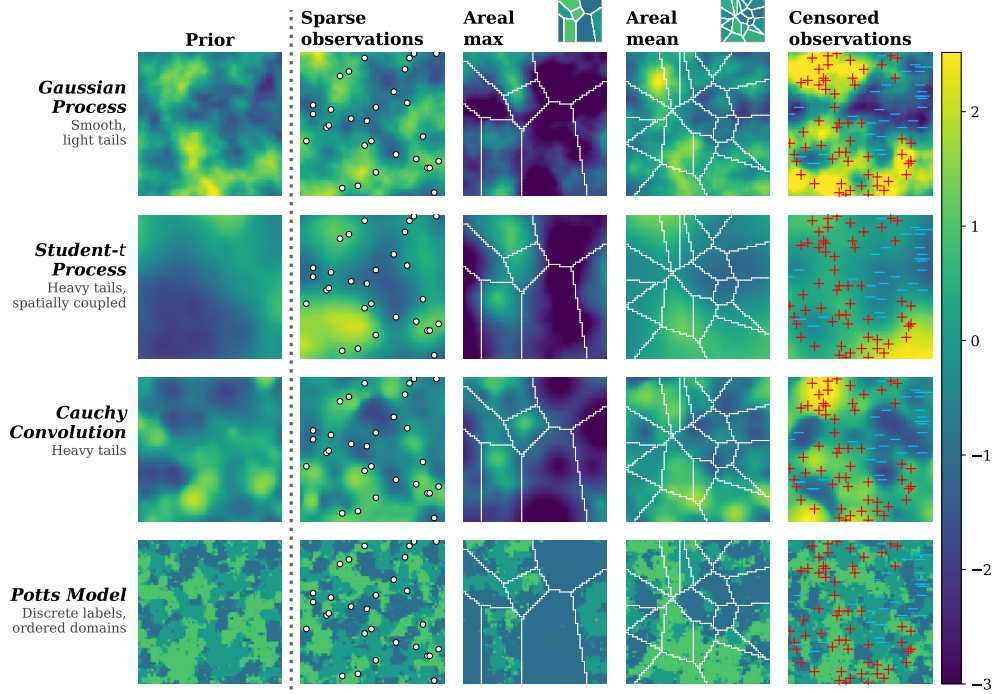


Figure 2: Posterior samples from four spatial process priors under four observation types via LATENTFLOW. **Rows:** Gaussian process (Matérn- $\frac{3}{2}$); Student- t process ($\nu = 4$); Cauchy convolution field; Potts model ($q = 3$). **Columns:** unconditional; 30 point observations; areal maxima (8 regions); areal means (20 regions); 60 censored observations (+/−). Thumbnails show areal structures. Sampling time: 3 – 4s per sample.

6.2 TEMPORAL PROCESSES

We next consider temporal processes. Examples include diffusion processes (e.g., Øksendal, 2003), and dynamical systems with random initial conditions or exogenous forcing (e.g., Law et al., 2015). This presents a different use case for LATENTFLOW: the object being conditioned is no longer a static field, but a trajectory generated by propagating latent randomness through a dynamical model.

Diffusion bridges. Diffusion bridges condition diffusion processes on a fixed terminal condition. They arise in many applications, including computational chemistry (Bolhuis et al., 2002), econometrics (Elerian et al., 2001), developmental dynamics (Wang et al., 2011), and shape analysis (Arnaudon et al., 2022). Except in a few tractable cases, the bridge drift depends on an unavailable transition density, motivating bespoke guided proposals and path-space MCMC methods (Delyon & Hu, 2006; Schauer et al., 2017), as well as recent neural approaches that approximate the bridge score or guidance field (Baker et al., 2024; Heng et al., 2025; Yang et al., 2025b). In Figure 3, we assess whether LATENTFLOW can generate diffusion bridges by acting on the exogenous simulation noise. In our case, exact endpoint conditioning is approximated by a narrow finite-width likelihood. Our method recovers the analytic OU bridge behaviour, and produces nonlinear cell-differentiation and FitzHugh–Nagumo paths consistent with rejection-sampled references. It also agrees qualitatively with existing methods (e.g., Yang et al., 2025b) in cases where no valid rejection samples were found after simulating the unconditioned process 100,000 times; see Figure 15 in Section G.3.

Trajectory-level conditioning. We next consider more general trajectory-level conditioning, where the information may involve observations, terminal events, or pathwise threshold constraints (e.g., Bolhuis & Swenson, 2021; Finzi et al., 2023; Grafke, 2025). Such constraints act on the realised trajectory after the latent randomness has been propagated through a nonlinear dynamical simulator, and need not reduce to a standard filtering, smoothing, or bridge-simulation problem. In Figure 4, we consider Lotka–Volterra dynamics (Lotka, 1925; Volterra, 1926), an SIR epidemic model (Kermack & McKendrick, 1927), and the Heston stochastic-volatility model (Heston, 1993). The resulting conditional paths satisfy the imposed trajectory-level information while retaining the characteristic dynamics of each model. Sampling remains very fast, taking roughly 1 second per sample for the SIR and Lotka–Volterra models, and roughly 3 seconds per sample for the Heston model.

6.3 SPATIO-TEMPORAL PROCESSES

We finally consider spatio-temporal models, which describe the random evolution of spatial fields. Such models arise in turbulent transport, environmental dispersion, materials science, excitable media, and phase separation (Majda & Kramer, 1999; Allen & Cahn, 1979; Nagumo et al., 1962). Conditioning them on sparse observations, future events, or aggregate space–time constraints is central to data assimilation, inverse modelling, and rare-event prediction (Dashti & Stuart, 2017; Cérou & Guyader, 2007; Cérou et al., 2012; Botev & L’Ecuyer, 2020).

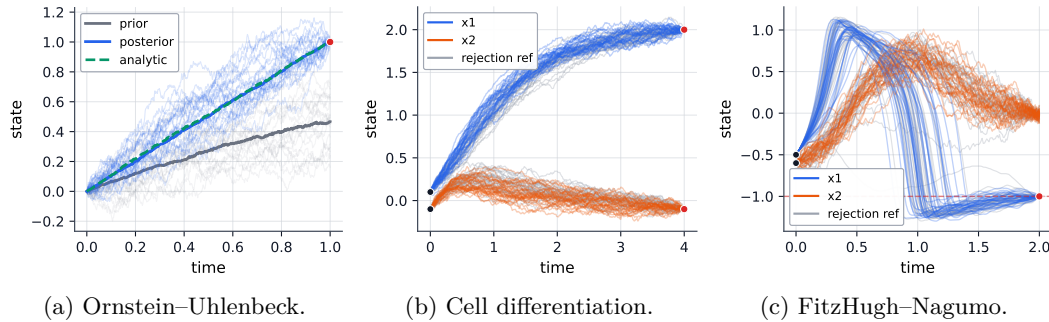


Figure 3: Diffusion bridges generated by LATENTFLOW. (a) An Ornstein–Uhlenbeck bridge. (b) A two-dimensional nonlinear cell-differentiation diffusion conditioned to reach a prescribed terminal state. (c) A FitzHugh–Nagumo diffusion conditioned on a rare transition between metastable states. Red markers indicate terminal targets, blue & orange curves indicate conditioned state components; grey curves indicate prior or rejection-reference trajectories.

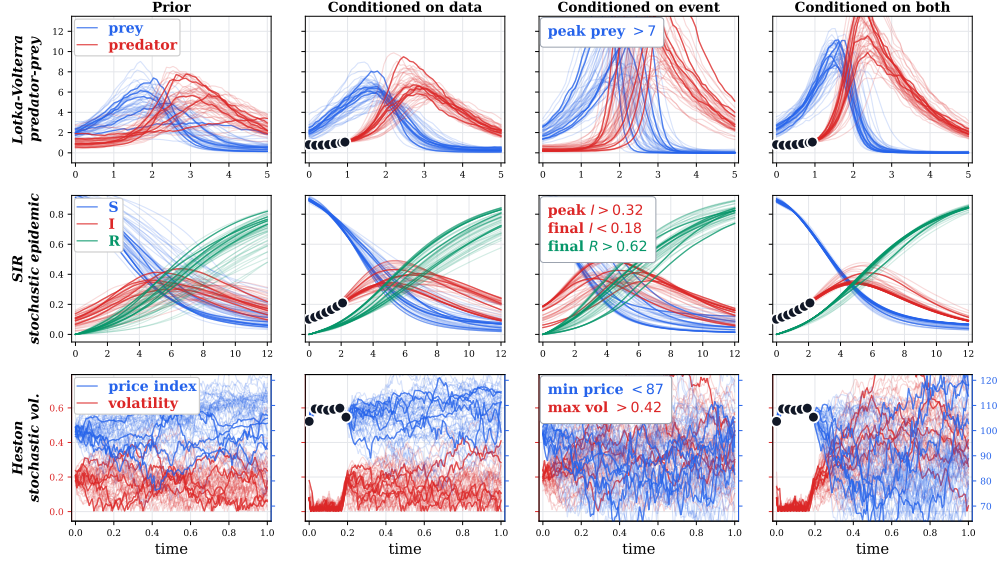


Figure 4: LATENTFLOW can condition SDE priors on data and rare events. **Rows:** Lotka–Volterra predator–prey dynamics, a stochastic SIR epidemic, and Heston stochastic volatility. **Columns:** the prior, the posterior given observations (black circles), the posterior given rare events: a prey boom, a fast growing but fast terminating epidemic, and a stress event (a market crash followed by a volatility spike). Sampling time: < 1 s for SIR and Lotka–Volterra; 3s Heston.

We consider three examples: a stochastic Allen–Cahn field for noisy phase separation, an advection–diffusion plume model, and a stochastic FitzHugh–Nagumo excitable-media model that produces travelling activation waves (FitzHugh, 1961; Nagumo et al., 1962; Allen & Cahn, 1979; Majda & Kramer, 1999). Representative samples are shown in Figure 5. Across all three models, LATENTFLOW produces fields that remain consistent with the prior dynamics while satisfying the imposed conditions.

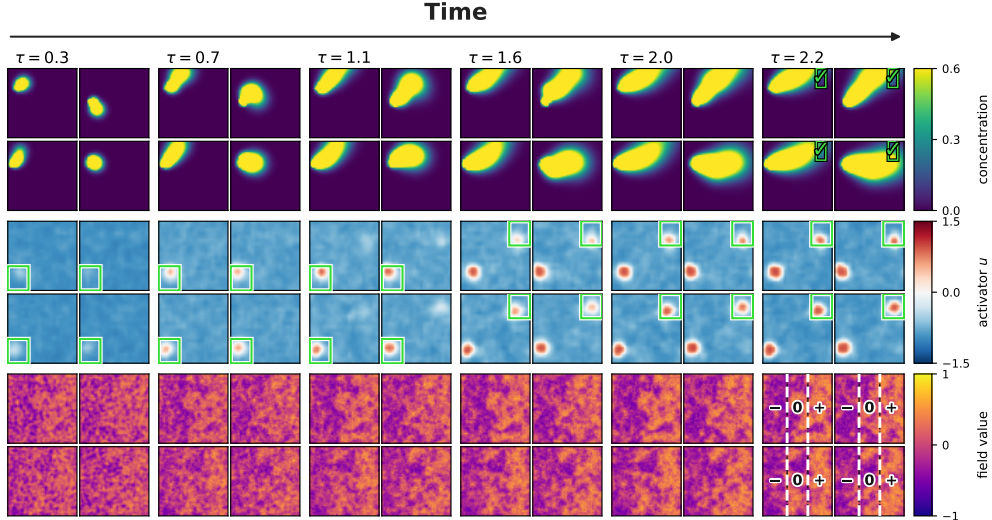


Figure 5: Posterior samples from three spatio-temporal systems using our LATENTFLOW: **(top)** a concentration plume conditioned on triggering a downstream detector evolving under advection–diffusion dynamics (160 latent dimensions); **(middle)** a FitzHugh–Nagumo excitable medium conditioned on sequential pacemaker firing (40K latent dimensions); and **(bottom)** an Allen–Cahn phase field conditioned on a prescribed phase front (122K latent dimensions). Each row shows four independent draws, over six time snapshots. Sampling time: 1s for the advection–diffusion; 5s for FitzHugh–Nagumo, and 2s for Allen–Cahn.

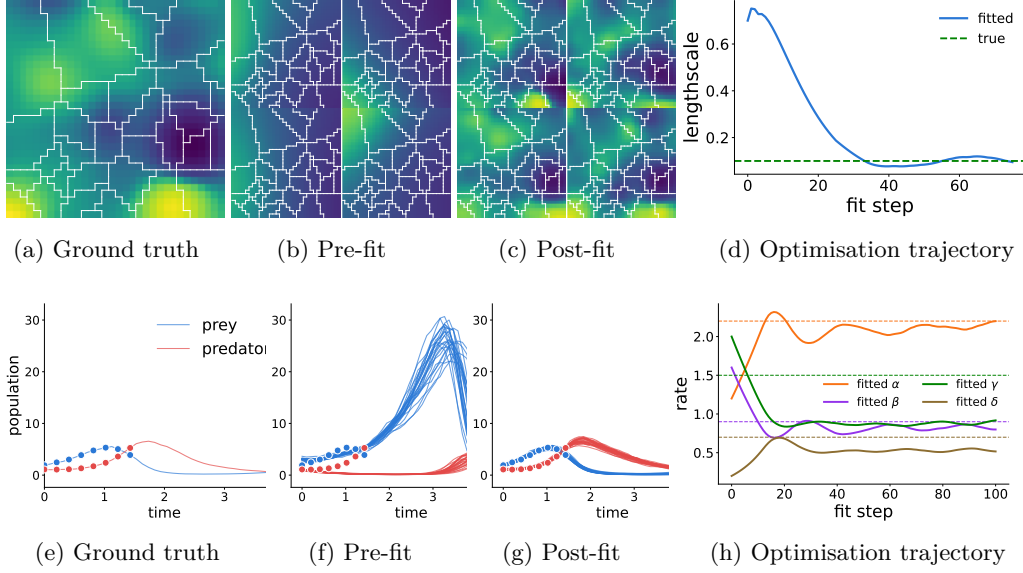


Figure 6: Guidance-aware parameter estimation recovers reasonable process hyperparameters, modulo identifiability. **Top:** Cauchy-convolution field. (a) Truth, observed only via the *maximum* value within each of $K = 32$ irregular regions. (b) Guided posterior samples under a mis-specified lengthscale ($\ell = 0.7$). (c) Samples after adapting the lengthscale against the regional-maximum condition. (d) Lengthscale trajectory converging to the true value ($\ell = 0.1$, dashed). **Bottom:** stochastic Lotka–Volterra process. (e) Truth, with dense early observations (dots). (f) Guided posterior samples under mis-specified rates. (g) Samples after jointly fitting all four rates against the same early-window condition. (h) Identifiable rates converge towards true values (dashed).

6.4 PARAMETER ESTIMATION

Finally, we assess the guidance-aware parameter adaptation framework introduced in [Section 4](#). We revisit two of the processes considered in earlier sections, and compare the results obtained with fixed parameters, with the hyperparameter selected after incorporating the full conditioning information. The results are shown in [Figure 6](#). Across both examples, our guidance-aware scheme recovers the ground truth, resulting in samples that more closely respect the target observations and constraints.

7 DISCUSSION

LATENTFLOW replaces the bespoke, model-by-model samplers that have previously been required for conditioning stochastic processes with a single, unified approach. The central idea is to move conditioning from process space to a latent innovation space, with the likelihood pulled back through the process generator and the resulting conditional law sampled using guided reverse-time dynamics. This gives an exact target-law construction, with practical approximation confined to three explicit and systematically reducible sources: finite terminal noising, Monte Carlo guidance, and time discretisation. This is a fundamental shift in capability. We expect the largest gains for rare, nonlinear, or global conditioning events under non-Gaussian and simulator-defined process priors, including extreme events, pathwise conditions, and constraints on SDE and SPDE driven models.

Several limitations remain. LATENTFLOW requires a *tractable* latent innovation representation of the discretised prior, and its performance can depend on the dimension, smoothness, and geometry of this representation. It also requires pointwise evaluation of the pulled-back likelihood, and ideally that the likelihood is differentiable through the generator. Discontinuous simulators, hard constraints, rare events, sharp likelihoods, multimodal conditionals, or poorly scaled latent parameterisations may therefore require relaxations, surrogates, derivative-free guidance, or larger Monte Carlo budgets.

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A EXAMPLES OF LATENT INNOVATION REPRESENTATIONS

This appendix contains examples of stochastic-process priors that admit the latent innovation representation in [Assumption 2.1](#), or its measure-theoretic extension; see [Remark B.1](#) for details. Throughout, $s_{1:m}$ denotes the discretisation inputs, $f_0 \in E_m$ denotes the finite-dimensional process realisation, and T_ϑ denotes a generator satisfying $f_0 = T_\vartheta(\xi_0)$.

In several examples, the model is most naturally written in terms of a native innovation variable ζ_0 , such as a uniform simulator seed, a Gamma scale variable, or a collection of non-Gaussian increments. To ensure all examples are consistent with the common latent form used by LATENTFLOW, we express this native innovation as a deterministic function of a standard Gaussian reference: $\zeta_0 = H_\vartheta(\xi_0)$, with $\xi_0 \sim \mathcal{N}(0, I)$. The generator in [Assumption 2.1](#) is then the composite map from the Gaussian innovation to the process realisation. When the native innovation is already Gaussian, H_ϑ is simply the identity map.

Table 1: Examples of stochastic-process priors and their latent innovation representations.

Model class	Innovation representation
<i>Gaussian, Student-t, and basis-expansion priors</i>	
Gaussian process	A finite-dimensional realisation $f_0 = (f(s_1), \dots, f(s_m))$ of a Gaussian process can be written as $f_0 = \mu_\vartheta + L_\vartheta \xi_0, \quad L_\vartheta L_\vartheta^\top = K_\vartheta,$ where $(K_\vartheta)_{ij} = k_\vartheta(s_i, s_j)$ is the kernel matrix, and $\xi_0 \sim \mathcal{N}(0, I_m)$. The innovation is given by $\xi_0 \in \mathbb{R}^m, \quad \xi_0 \sim \mathcal{N}(0, I_m).$ The generator is given by $T_\vartheta(\xi_0) = \mu_\vartheta + L_\vartheta \xi_0.$ This is the affine whitening map used in finite-dimensional Gaussian process inference.
Student-t process	A finite-dimensional realisation $f_0 = (f(s_1), \dots, f(s_m))$ of a Student-t process can be written as $f_0 = \mu_\vartheta + \omega^{-1/2} L_\vartheta v, \quad L_\vartheta L_\vartheta^\top = K_\vartheta,$ where $(K_\vartheta)_{ij} = k_\vartheta(s_i, s_j)$ is the scale matrix induced by the Student-t process kernel k_ϑ, and v and ω are independent, with $v \sim \mathcal{N}(0, I_m)$ and $\omega \sim \text{Gamma}(\nu/2, \nu/2)$ in the shape-rate parameterisation. The natural innovation is given by $\zeta_0 = (v, \omega) \in \mathbb{R}^{m+1}.$ Let $w = \Phi_N^{-1}(F_{\Gamma, \nu}(\omega))$, where $F_{\Gamma, \nu}$ is the distribution function of $\text{Gamma}(\nu/2, \nu/2)$, and Φ_N is the standard normal distribution function. The Gaussian innovation is then given by $\xi_0 = (v, w) \in \mathbb{R}^{m+1}, \quad \xi_0 \sim \mathcal{N}(0, I_{m+1}).$ The corresponding generator is given by $T_\vartheta(\xi_0) = \mu_\vartheta + F_{\Gamma, \nu}^{-1}(\Phi_N(w))^{-1/2} L_\vartheta v.$ This gives a heavy-tailed analogue of the Gaussian process while retaining an explicit scale-mixture innovation map.

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Model class	Innovation representation
Finite-basis, random-feature, or pathwise GP approximation	A truncated Karhunen–Loève expansion, finite basis expansion, random-feature approximation, or pathwise Gaussian process approximation can be written as $f_0 = \mu_\vartheta + \Phi_\vartheta \xi_0,$ where $\Phi_\vartheta \in \mathbb{R}^{m \times q}$ contains the basis or feature evaluations at $s_{1:m}$, and $\xi_0 \sim \mathcal{N}(0, I_q)$. For example, $(\Phi_\vartheta)_{ik} = \sqrt{\lambda_{\vartheta,k}} \phi_{\vartheta,k}(s_i)$ for a truncated Karhunen–Loève expansion. The innovation is given by $\xi_0 \in \mathbb{R}^q, \quad \xi_0 \sim \mathcal{N}(0, I_q).$ The generator is given by $T_\vartheta(\xi_0) = \mu_\vartheta + \Phi_\vartheta \xi_0.$ The innovation dimension q need not equal the discretisation dimension m. When $q < m$, the induced prior on E_m is typically supported on a low-dimensional subspace unless an additional residual or nugget term is included. More elaborate pathwise Gaussian process constructions fit the same form after augmenting ξ_0 with auxiliary Gaussian variables.
Spectral random field	A finite spectral random field can be written as $f(s) = \mu_\vartheta(s) + \sum_{k=1}^q A_{\vartheta,k} (a_k \cos(\omega_k^\top s) + b_k \sin(\omega_k^\top s)),$ where $A_{\vartheta,k}$ are spectral amplitudes, ω_k are frequencies, and $a_k, b_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. Here the frequencies ω_k are treated as fixed, or as part of ϑ. If the frequencies are themselves random, they can be appended to the innovation variable. The innovation is given by $\xi_0 = (a_1, b_1, \dots, a_q, b_q) \in \mathbb{R}^{2q}, \quad \xi_0 \sim \mathcal{N}(0, I_{2q}).$ The generator is given by $(T_\vartheta(\xi_0))_i = \mu_\vartheta(s_i) + \sum_{k=1}^q A_{\vartheta,k} (a_k \cos(\omega_k^\top s_i) + b_k \sin(\omega_k^\top s_i)),$ for components $i = 1, \dots, m$.
Gaussian Markov random field, or SPDE-defined Matérn field	A Gaussian Markov random field, including an SPDE-defined Matérn field after finite-element discretisation, can be written as $R_\vartheta^\top R_\vartheta = Q_\vartheta, \quad f_0 = \mu_\vartheta + A_\vartheta R_\vartheta^{-1} \xi_0,$ where $x_\vartheta = R_\vartheta^{-1} \xi_0 \in \mathbb{R}^q$ is a vector of GMRF or finite-element coefficients, $Q_\vartheta \succ 0$ is its sparse precision matrix, $A_\vartheta : \mathbb{R}^q \rightarrow E_m$ maps the coefficients to the requested process values, and $\xi_0 \sim \mathcal{N}(0, I_q)$. For an SPDE-defined Matérn field, Q_ϑ is the sparse precision matrix of the finite-element coefficient vector, and A_ϑ is the corresponding finite-element evaluation or projection matrix. The innovation is given by $\xi_0 \in \mathbb{R}^q, \quad \xi_0 \sim \mathcal{N}(0, I_q).$ The generator is given by $T_\vartheta(\xi_0) = \mu_\vartheta + A_\vartheta R_\vartheta^{-1} \xi_0.$ In practice, one solves a sparse triangular system rather than forming R_ϑ^{-1}. This includes proper areal Gaussian Markov random fields such as proper CAR and Leroux-type models, as well as SPDE-defined Matérn fields. Intrinsic CAR or intrinsic GMRF precision matrices are singular and instead require an identifying constraint, a reduced-coordinate factorisation on the constrained subspace, or an explicitly specified generalised-inverse construction.

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Model class	Innovation representation
<i>Intensity and convolution-field priors</i>	
Log-Gaussian Cox intensity	A discretised log-Gaussian Cox process intensity can be written as $f_0 = \exp(\mu_\vartheta + L_\vartheta \xi_0), \quad L_\vartheta L_\vartheta^\top = K_\vartheta,$ where $f_0 = (\lambda(s_1), \dots, \lambda(s_m))$, $g_0 = \mu_\vartheta + L_\vartheta \xi_0$ is the discretised latent Gaussian process, $\lambda(s_i) = \exp(g_0(s_i))$, and $\xi_0 \sim \mathcal{N}(0, I_m)$. The innovation is given by $\xi_0 \in \mathbb{R}^m, \quad \xi_0 \sim \mathcal{N}(0, I_m).$ The generator is given by $T_\vartheta(\xi_0) = \exp(\mu_\vartheta + L_\vartheta \xi_0),$ where the exponential is applied componentwise.
Lévy-driven or convolution random field	A finite Lévy-driven or stable-convolution random field can be written as $f(s_i) = \sum_{k=1}^q G_\vartheta(s_i - c_k) J_k, \quad i = 1, \dots, m,$ where G_ϑ is a convolution kernel, c_k are support or quadrature locations, and $J_k \sim \nu_{\vartheta,k}$ are increments of an independently scattered random measure. The natural innovation is given by $\zeta_0 = (J_1, \dots, J_q).$ Let $H_{\vartheta,k}$ be a map such that $J_k = H_{\vartheta,k}(w_k), \quad w_k \sim \mathcal{N}(0, I).$ The Gaussian innovation is then given by $\xi_0 = (w_1, \dots, w_q).$ The corresponding generator is given by $(T_\vartheta(\xi_0))_i = \sum_{k=1}^q G_\vartheta(s_i - c_k) H_{\vartheta,k}(w_k), \quad i = 1, \dots, m.$ This gives a non-Gaussian random-field prior, including Cauchy, normal-inverse Gaussian (NIG), stable, or other heavy-tailed convolution fields, through independently scattered random-measure increments.

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Model class	Innovation representation
<i>Discrete and extreme-value priors</i>	
Potts or discrete Markov random field	A Potts or discrete Markov random field on sites $s_{1:m}$ can be written as $\mathbb{P}_\vartheta(f_0 = x) \propto \exp\left(\sum_{i=1}^m h_{\vartheta,i}(x_i) + \sum_{(i,j) \in \mathcal{E}} \beta_{\vartheta,ij} \mathbf{1}(x_i = x_j)\right),$ where $x = (x_1, \dots, x_m) \in \{1, \dots, K\}^m$, $x_i = f_0(s_i)$, $\mathcal{E}$ is the neighbourhood graph on $s_{1:m}$, $h_{\vartheta,i}$ are site potentials, and $\beta_{\vartheta,ij}$ are interaction parameters. The natural innovation can be represented by a simulator seed $\zeta_0 = u \in (0, 1)^q, \quad u \sim \text{Unif}((0, 1)^q),$ together with an exact sampler $\mathcal{S}_\vartheta$ such that $f_0 = \mathcal{S}_\vartheta(u) \sim \mathbb{P}_\vartheta$. If $\mathcal{S}_\vartheta$ is an approximate MCMC or sequential sampler, then the corresponding T_ϑ represents the algorithmic approximation rather than the exact Potts law. Let $\xi_0 = \Phi_N^{-1}(u)$, where Φ_N is the standard normal distribution function, applied componentwise. The Gaussian innovation is then given by $\xi_0 \in \mathbb{R}^q, \quad \xi_0 \sim \mathcal{N}(0, I_q).$ The corresponding generator is given by $T_\vartheta(\xi_0) = \mathcal{S}_\vartheta(\Phi_N(\xi_0)).$ Exact Potts fields are discrete; gradient-based guidance therefore requires a relaxation, surrogate, or derivative-free guidance estimator.
Max-stable process	A finite spectral truncation of a max-stable process can be written as $f(s_i) = \max_{1 \leq k \leq K} P_k W_{\vartheta,k}(s_i), \quad P_k = \Gamma_k^{-1}, \quad \Gamma_k = \sum_{\ell=1}^k E_\ell,$ where $E_k \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1)$, the spectral functions $W_{\vartheta,k}$ are independent and identically distributed, and the sequence $(W_{\vartheta,k})_{k \geq 1}$ is independent of $(E_k)_{k \geq 1}$. The spectral functions satisfy the usual normalisation $\mathbb{E}[W_{\vartheta,k}(s)] = 1$ for each s. For finite K, this is a spectral truncation of the corresponding infinite max-stable representation. The natural innovation is given by $\zeta_0 = (E_{1:K}, v_{1:K}),$ where v_k generates the spectral function $W_{\vartheta,k}$. Let $w_k^E = \Phi_N^{-1}(\exp(-E_k)),$ so that, equivalently, $E_k = -\log \Phi_N(w_k^E), \quad w_k^E \sim \mathcal{N}(0, 1).$ If the spectral innovation can be Gaussianised as $v_k = H_\vartheta^W(w_k^W), \quad w_k^W \sim \mathcal{N}(0, I),$ then the Gaussian innovation is given by $\xi_0 = (w_1^E, \dots, w_K^E, w_1^W, \dots, w_K^W).$ All variables w_k^E and w_k^W are taken to be mutually independent across k and across the two innovation blocks. The corresponding generator is given by $(T_\vartheta(\xi_0))_i = \max_{1 \leq k \leq K} \Gamma_k^{-1} W_{\vartheta}(s_i; H_\vartheta^W(w_k^W)), \quad i = 1, \dots, m,$ where $\Gamma_k = \sum_{\ell=1}^k -\log \Phi_N(w_\ell^E)$. The maximum is non-smooth, so smooth relaxations may be required for gradient-based guidance.

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Model class	Innovation representation
<i>Dynamical and spatiotemporal priors</i>	
State-space model or stochastic recurrence	A state-space model or stochastic recurrence can be written as $x_0 = \Psi_{\vartheta,0}(\eta_0),$ $x_n = \Psi_{\vartheta,n}(x_{n-1}, \eta_n), \quad n = 1, \dots, N,$ where $\eta_n \stackrel{\text{ind}}{\sim} r_n$ are model innovations and $\Psi_{\vartheta,n}$ is the one-step simulator. The natural innovation is given by $\zeta_0 = (\eta_0, \dots, \eta_N).$ Let $H_{\vartheta,n}$ be a map such that $\eta_n = H_{\vartheta,n}(w_n), \quad w_n \sim \mathcal{N}(0, I).$ The Gaussian innovation is then given by $\xi_0 = (w_0, \dots, w_N).$ The corresponding generator is given by $x_0 = \Psi_{\vartheta,0}(H_{\vartheta,0}(w_0)),$ $x_n = \Psi_{\vartheta,n}(x_{n-1}, H_{\vartheta,n}(w_n)), \quad n = 1, \dots, N,$ with output $T_{\vartheta}(\xi_0) = (x_0, \dots, x_N).$ When the original innovations are already Gaussian, $H_{\vartheta,n}$ is the identity map. For discrete or mixed innovations, the maps $H_{\vartheta,n}$ may be nonsmooth, in which case gradient-based guidance again requires a relaxation, surrogate, or derivative-free estimator. Linear Gaussian and nonlinear Gaussian state-space models correspond to particular choices of $\Psi_{\vartheta,n}$.
Diffusion process	A diffusion process satisfying $dx_{\tau} = b_{\vartheta}(\tau, x_{\tau}) d\tau + \Sigma_{\vartheta}(\tau, x_{\tau}) dw_{\tau}$ can be discretised by Euler–Maruyama as $x_{n+1} = x_n + b_{\vartheta}(\tau_n, x_n)\Delta_n + \Sigma_{\vartheta}(\tau_n, x_n)\sqrt{\Delta_n}\eta_n,$ where $\Delta_n = \tau_{n+1} - \tau_n$, $\eta_n \stackrel{\text{ind}}{\sim} \mathcal{N}(0, I)$, and $\Sigma_{\vartheta}(\tau_n, x_n)$ maps Gaussian innovations into diffusion increments. The innovation is given by $\xi_0 = (\eta_{-1}, \eta_0, \dots, \eta_{N-1}),$ where $\eta_{-1} \sim \mathcal{N}(0, I)$ generates the initial state if x_0 is random, through an initial-state map $x_0 = \chi_{\vartheta,0}(\eta_{-1})$, and is omitted otherwise. The generator is given by applying the Euler–Maruyama recursion above and returning $T_{\vartheta}(\xi_0) = (x_0, \dots, x_N).$ More generally, higher-order numerical solvers correspond to different choices of T_{ϑ}.

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Model class	Innovation representation
Jump diffusion, Lévy-driven SDE, or Lévy process	A jump diffusion or Lévy-driven SDE satisfying $dx_\tau = b_\vartheta(\tau, x_\tau) d\tau + \Sigma_\vartheta(\tau, x_\tau) dW_\tau + dj_\tau$ can be discretised as $x_{n+1} = x_n + b_\vartheta(\tau_n, x_n)\Delta_n + \Sigma_\vartheta(\tau_n, x_n)\sqrt{\Delta_n}\eta_n + \ell_n,$ where $\Delta_n = \tau_{n+1} - \tau_n$, $\eta_n \stackrel{\text{ind}}{\sim} \mathcal{N}(0, I)$, and $\ell_n \sim \nu_{\vartheta, \Delta_n}$ is the jump or Lévy increment over $[\tau_n, \tau_{n+1}]$. The natural innovation is given by $\xi_0 = (\eta_{-1}, \eta_0, \dots, \eta_{N-1}, \ell_0, \dots, \ell_{N-1}),$ where $\eta_{-1} \sim \mathcal{N}(0, I)$ generates the initial state if x_0 is random, through an initial-state map $x_0 = \chi_{\vartheta, 0}(\eta_{-1})$, and is omitted otherwise. Let H_{ϑ, Δ_n} be a map such that $\ell_n = H_{\vartheta, \Delta_n}(w_n), \quad w_n \sim \mathcal{N}(0, I).$ The Gaussian innovation is then given by $\xi_0 = (\eta_{-1}, \eta_0, \dots, \eta_{N-1}, w_0, \dots, w_{N-1}).$ The corresponding generator is given by applying the jump–Euler recursion with $\ell_n = H_{\vartheta, \Delta_n}(w_n)$ and returning $T_\vartheta(\xi_0) = (x_0, \dots, x_N).$ A Lévy process or subordinator is obtained as the special case with no state-dependent drift or diffusion term and additive Lévy increments.
Time-discretised SPDE	A spatially discretised SPDE of the form $du_\tau = \mathcal{A}_\vartheta(u_\tau) d\tau + \mathcal{B}_\vartheta(u_\tau) dW_\tau$ can be time-discretised as $u_{n+1} = \Psi_{\vartheta, n}(u_n, \eta_n), \quad n = 0, \dots, N-1,$ where $\eta_n \stackrel{\text{ind}}{\sim} \mathcal{N}(0, I)$ are finite-dimensional Gaussian noise innovations and $\Psi_{\vartheta, n}$ is the numerical SPDE time-stepper. The innovation is given by $\xi_0 = (\eta_{-1}, \eta_0, \dots, \eta_{N-1}),$ where $\eta_{-1} \sim \mathcal{N}(0, I)$ generates the initial state if u_0 is random, through an initial-state map $u_0 = \chi_{\vartheta, 0}(\eta_{-1})$, and is omitted otherwise. The generator is given by applying the numerical SPDE solver above and returning the discretised space-time field.

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Model class	Innovation representation
<i>Implicit and simulator-defined priors</i>	
Neural or deep implicit process	A neural or deep implicit process can be written as $f(s_i) = G_\vartheta(s_i, h, \epsilon_i), \quad i = 1, \dots, m,$ where G_ϑ is a neural decoder, h is a global latent variable, and ϵ_i are optional local noise variables. The innovation is given by $\xi_0 = (h, \epsilon_1, \dots, \epsilon_m),$ with a simple reference law, typically standard Gaussian. The generator is given by $(T_\vartheta(\xi_0))_i = G_\vartheta(s_i, h, \epsilon_i), \quad i = 1, \dots, m.$ If the decoder is differentiable, $D_\xi T_\vartheta$ is available by automatic differentiation.
Black-box simulator	A black-box simulator can be written as $f_0 = \mathcal{S}_\vartheta(s_{1:m}; u),$ where $\mathcal{S}_\vartheta$ is the simulator and $u \sim \text{Unif}((0, 1)^q)$ is the simulator random seed. The natural innovation is given by $\zeta_0 = u \in (0, 1)^q.$ Let $\xi_0 = \Phi_N^{-1}(u)$, where Φ_N is the standard normal distribution function, applied componentwise. Then $\xi_0 \sim \mathcal{N}(0, I_q).$ The corresponding generator is given by $T_\vartheta(\xi_0) = \mathcal{S}_\vartheta(s_{1:m}; \Phi_N(\xi_0)).$ If $\mathcal{S}_\vartheta$ is not differentiable, guidance requires a smooth surrogate, relaxation, or derivative-free estimator.

B PROOFS FOR THE LATENT CONDITIONING PRINCIPLE

Proof of Proposition 2.2. Let $\xi_0 \sim \rho_\vartheta(\cdot \mid \mathcal{C})$. Then, for any measurable $A \subseteq E_m$, we have that

$$\begin{aligned} \mathbb{P}(T_\vartheta(\xi_0) \in A) &= \frac{1}{Z_\vartheta} \int_{\mathbb{R}^{m_\xi}} \mathbb{1}_{\{T_\vartheta(\xi_0) \in A\}} L_C(T_\vartheta(\xi_0)) r(\xi_0) d\xi_0 \\ &= \frac{1}{Z_\vartheta} \int_A L_C(f_0) p_\vartheta(f_0) df_0 = \int_A \pi_\vartheta(f_0 \mid \mathcal{C}) df_0. \end{aligned}$$

The third equality follows from the pushforward relation $T_{\vartheta\#}(r(\xi_0) d\xi_0) = p_\vartheta(f_0) df_0$, applied to the nonnegative measurable function $g(f_0) = \mathbf{1}_A(f_0) L_C(f_0)$. Hence $T_\vartheta(\xi_0)$ has density $\pi_\vartheta(\cdot \mid \mathcal{C})$, as required. $\square$

Remark B.1. We use density notation in the main text to simplify the exposition. Our approach, however, admits a much more general measure-theoretic formulation. Let P_ϑ denote the prior law of f_0 . Assume that $L_C : E_m \rightarrow [0, \infty)$ is measurable. The conditional law on E_m is then

$$\Pi_\vartheta(df_0 \mid \mathcal{C}) = \frac{L_C(f_0) P_\vartheta(df_0)}{Z_\vartheta}, \quad Z_\vartheta := \int L_C(f_0) P_\vartheta(df_0). \quad (8)$$

Our assumption now reads as follows. There exist a latent dimension m_ξ , a reference probability measure R on $\mathbb{R}^{m_\xi}$, and a measurable map $T_\vartheta : \mathbb{R}^{m_\xi} \rightarrow E_m$ such that $P_\vartheta = (T_\vartheta)_\# R$. Equivalently, if $\xi_0 \sim R$, then $f_0 = T_\vartheta(\xi_0) \sim P_\vartheta$. The corresponding latent target is given by

$$\rho_\vartheta(d\xi \mid \mathcal{C}) = \frac{L_C(T_\vartheta(\xi)) R(d\xi)}{Z_\vartheta}, \quad Z_\vartheta = \int L_C(T_\vartheta(\xi)) R(d\xi). \quad (9)$$

We emphasise that the values of Z_ϑ as defined in (8) and (9) coincide, since $P_\vartheta = T_{\vartheta\#} R$ implies $\int L_C(f_0) P_\vartheta(df_0) = \int L_C(T_\vartheta(\xi)) R(d\xi)$ by definition of the pushforward. Now, for any

measurable set $A \subseteq E_m$, we have that

$$\begin{aligned} T_{\vartheta\#}\rho_{\vartheta}(\cdot | \mathcal{C})(A) &= \int \mathbb{1}_{\{T_{\vartheta}(\xi) \in A\}} \frac{L_{\mathcal{C}}(T_{\vartheta}(\xi))R(d\xi)}{Z_{\vartheta}} \\ &= \frac{1}{Z_{\vartheta}} \int_A L_{\mathcal{C}}(f_0)P_{\vartheta}(df_0) = \Pi_{\vartheta}(A | \mathcal{C}). \end{aligned}$$

It follows that $T_{\vartheta\#}\rho_{\vartheta}(\cdot | \mathcal{C}) = \Pi_{\vartheta}(\cdot | \mathcal{C})$. This is the measure-theoretic form of [Proposition 2.2](#). The density-based statement in the main text is recovered when P_{ϑ} admits a density p_{ϑ} . This more general argument also covers singular and discrete laws.

Our next result establishes that the latent representation also preserves the relative-entropy cost of conditioning.

Proposition B.2. *Suppose that Assumption 2.1 holds and that $0 < Z_{\vartheta} < \infty$. Then, in the extended sense,*

$$\text{KL}(\rho_{\vartheta}(\cdot | \mathcal{C}) \| r) = \text{KL}(\pi_{\vartheta}(\cdot | \mathcal{C}) \| p_{\vartheta}).$$

Proof. By construction, we have

$$\frac{\rho_{\vartheta}(\xi_0 | \mathcal{C})}{r(\xi_0)} = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi_0)}{Z_{\vartheta}} = \frac{L_{\mathcal{C}}(T_{\vartheta}(\xi_0))}{Z_{\vartheta}} \quad r\text{-a.e.} \quad (10)$$

Similarly, due to (1), it holds that

$$\frac{\pi_{\vartheta}(f_0 | \mathcal{C})}{p_{\vartheta}(f_0)} = \frac{L_{\mathcal{C}}(f_0)}{Z_{\vartheta}} \quad p_{\vartheta}\text{-a.e.} \quad (11)$$

Let $\ell(f_0) = \log\left(\frac{L_{\mathcal{C}}(f_0)}{Z_{\vartheta}}\right)$, with the usual extended-value convention. Using (10), we obtain

$$\text{KL}(\rho_{\vartheta}(\cdot | \mathcal{C}) \| r) = \int_{\mathbb{R}^{m_{\xi}}} \ell(T_{\vartheta}(\xi_0))\rho_{\vartheta}(\xi_0 | \mathcal{C}) d\xi_0.$$

By [Proposition 2.2](#), pushing $\rho_{\vartheta}(\cdot | \mathcal{C})$ forward through T_{ϑ} gives $\pi_{\vartheta}(\cdot | \mathcal{C})$. We therefore have,

$$\int_{\mathbb{R}^{m_{\xi}}} \ell(T_{\vartheta}(\xi_0))\rho_{\vartheta}(\xi_0 | \mathcal{C}) d\xi_0 = \int_{E_m} \ell(f_0)\pi_{\vartheta}(f_0 | \mathcal{C}) df_0. \quad (12)$$

Finally, using (11), the right-hand side of (12) is exactly $\text{KL}(\pi_{\vartheta}(\cdot | \mathcal{C}) \| p_{\vartheta})$. This proves the identity. $\square$

This result suggests that the latent construction is not just a convenient sampling device. Since the additional likelihood depends on the latent variable only through the generated process value $T_{\vartheta}(\xi_0)$, the relative-entropy deformation from the prior p_{ϑ} to the conditional $\pi_{\vartheta}(\cdot | \mathcal{C})$ is exactly matched by the deformation from the reference r to the latent conditional $\rho_{\vartheta}(\cdot | \mathcal{C})$. In contrast, for a generic latent density η on $\mathbb{R}^{m_{\xi}}$, if p_{η} denotes the density of $T_{\vartheta}(\xi_0)$ when $\xi_0 \sim \eta$, then the data-processing inequality only gives $\text{KL}(p_{\eta} \| p_{\vartheta}) \leq \text{KL}(\eta \| r)$. Thus the latent target introduces no additional information beyond the output-level conditioning likelihood.

C EXACTNESS OF THE LATENT DYNAMICS

We now fix ϑ and $\mathcal{C}$, and write $T = T_{\vartheta}$, $\Lambda = \Lambda_{\vartheta, \mathcal{C}}$, $Z = Z_{\vartheta}$, and $\rho_0(\xi_0) = \rho_{\vartheta}(\xi_0 | \mathcal{C})$. We also write $r(\xi) = \mathcal{N}(\xi; 0, I_{m_{\xi}})$.

Recall that the VP noising schedule is specified by a locally integrable function $\beta : [0, 1) \rightarrow (0, \infty)$. We write

$$B(t) = \int_0^t \beta(s) ds, \quad \alpha(t) = \exp\left\{-\frac{1}{2}B(t)\right\}, \quad \sigma(t)^2 = 1 - \alpha(t)^2.$$

We now allow either a finite-noising horizon, where $B(1) := \lim_{t \uparrow 1} B(t) < \infty$, or an ideal infinite-noising horizon, where $B(1) = \infty$ and hence $\alpha(t) \downarrow 0$ as $t \uparrow 1$. For $t \in [0, 1)$, the VP bridge is

$$\xi_t = \alpha(t)\xi_0 + \sigma(t)\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_{m_\varepsilon}), \quad (13)$$

where ε is independent of ξ_0 . We write ρ_t for the density of the noised latent variable obtained by drawing $\xi_0 \sim \rho_0$ and applying (13). Equivalently,

$$\rho_t(u) = \frac{\psi(t, u)r(u)}{Z}, \quad \psi(t, u) = \mathbb{E}_{\xi_0 \sim r} [\Lambda(\xi_0) \mid \xi_t = u].$$

where the conditional expectation is taken under the reference bridge with $\xi_0 \sim r$. Finally, assuming $\psi(t, \cdot)$ is differentiable, we write $g(t, u) = \nabla_u \log \psi(t, u)$ for the latent guidance field.

In Section 3.2, we formally identified the SDE associated with the guided latent marginals $(\rho_t)_{0 \leq t < 1}$. We now recall these dynamics using notation that is convenient for the exactness statements below. The forward noising SDE is given by

$$d\xi_t = -\frac{1}{2}\beta(t)\xi_t dt + \sqrt{\beta(t)} dw_t, \quad \xi_0 \sim \rho_0, \quad 0 \leq t < 1,$$

where $w = (w_t)_{0 \leq t < 1}$ is a standard $\mathbb{R}^{m_\varepsilon}$ -valued Brownian motion. Fix $0 \leq t_0 < t_1 < 1$, and set $\tau_s = t_1 - s$, for $0 \leq s \leq t_1 - t_0$. The reverse-time SDE is then given by

$$d\xi_s^\leftarrow = \left[\beta(\tau_s)g(\tau_s, \xi_s^\leftarrow) - \frac{1}{2}\beta(\tau_s)\xi_s^\leftarrow \right] ds + \sqrt{\beta(\tau_s)} d\bar{w}_s, \quad \xi_0^\leftarrow \sim \rho_{t_1}, \quad (14)$$

where $\bar{w} = (\bar{w}_s)_{0 \leq s \leq t_1 - t_0}$ is a standard Brownian motion in reverse time. The same family of marginals can also be generated by a deterministic process known as the probability-flow ODE (e.g., Song et al., 2021). In forward noising time, this ODE is given by

$$\frac{du_t}{dt} = -\frac{1}{2}\beta(t)g(t, u_t), \quad u_0 \sim \rho_0, \quad 0 \leq t < 1. \quad (15)$$

For sampling, we again use the reverse-time parametrisation $u_s^\leftarrow = u_{\tau_s}$, with τ_s as defined above. In particular, the reverse-time probability-flow ODE is

$$\frac{du_s^\leftarrow}{ds} = \frac{1}{2}\beta(\tau_s)g(\tau_s, u_s^\leftarrow), \quad u_0^\leftarrow \sim \rho_{t_1}. \quad (16)$$

Assumption C.1. For every compact interval $[t_0, t_1] \subset (0, 1)$, ρ_t is a strictly positive Lebesgue density for all $t \in [t_0, t_1]$. The map $(t, u) \mapsto \nabla_u \log \rho_t(u)$ is sufficiently regular that the reverse-time SDE (14) and the probability-flow ODE (16) are well posed on the corresponding time intervals, and their associated Fokker–Planck and continuity equations have unique density-valued solutions.

Theorem C.2. Suppose that Assumption 2.1 holds, that $r = \mathcal{N}(0, I_{m_\varepsilon})$, that $0 < Z < \infty$, and that Assumption C.1 holds. Fix $0 < t_0 < t_1 < 1$, and define $\tau_s = t_1 - s$ for $0 \leq s \leq t_1 - t_0$. Let $\xi_0^\leftarrow \sim \rho_{t_1}$, and let $(\xi_s^\leftarrow)_{0 \leq s \leq t_1 - t_0}$ solve the reverse-time SDE (14). Then $\xi_s^\leftarrow \sim \rho_{\tau_s}$ for all $0 \leq s \leq t_1 - t_0$. In particular, $\xi_{t_1 - t_0}^\leftarrow \sim \rho_{t_0}$. If the reverse SDE is well defined down to $t_0 = 0$, then

$$\xi_{t_1}^\leftarrow \sim \rho_0 = \rho_\vartheta(\cdot \mid \mathcal{C}), \quad T_\vartheta(\xi_{t_1}^\leftarrow) \sim \pi_\vartheta(\cdot \mid \mathcal{C}).$$

Proof. The proof is standard. Let $b_t(u) = -\frac{1}{2}\beta(t)u$ denote the drift of the forward VP SDE. By the standard time-reversal formula for diffusions (e.g., Anderson, 1982), the reverse-time diffusion with clock $\tau_s = t_1 - s$ has drift given by

$$\tilde{b}_s(u) = -b_{\tau_s}(u) + \beta(\tau_s)\nabla_u \log \rho_{\tau_s}(u). \quad (17)$$

Moreover, since $\rho_t(u) = Z^{-1}\psi(t, u)r(u)$ and $r(u) = \mathcal{N}(u; 0, I_{m_\varepsilon})$, it follows straightforwardly that

$$\nabla_u \log \rho_t(u) = \nabla_u \log \psi(t, u) - u = g(t, u) - u. \quad (18)$$

We thus have, substituting (18) into (17), that

$$\tilde{b}_s(u) = \frac{1}{2}\beta(\tau_s)u + \beta(\tau_s)(g(\tau_s, u) - u) = \beta(\tau_s)g(\tau_s, u) - \frac{1}{2}\beta(\tau_s)u.$$

This is exactly the drift in (14). Equivalently, if $q_s(u) = \rho_{\tau_s}(u)$, then q_s solves the Fokker-Planck equation associated with (14). Indeed,

$$\partial_s q_s = -\partial_t \rho_t|_{t=\tau_s} = \nabla \cdot (b_{\tau_s} \rho_{\tau_s}) - \frac{1}{2} \beta(\tau_s) \Delta \rho_{\tau_s} = -\nabla \cdot (\tilde{b}_s q_s) + \frac{1}{2} \beta(\tau_s) \Delta q_s.$$

By uniqueness in Assumption C.1, the law of $\xi_s^\leftarrow$ has density $q_s = \rho_{\tau_s}$. The claims at $t_0 = 0$ follow because $\rho_0 = \rho_\vartheta(\cdot | \mathcal{C})$. Finally, the process-space statement follows from Proposition 2.2. $\square$

Theorem C.3. *Suppose that Assumption 2.1 holds, that $r = \mathcal{N}(0, I_{m_\xi})$, that $0 < Z < \infty$, and that Assumption C.1 holds. Fix $0 < t_0 < t_1 < 1$, and define $\tau_s = t_1 - s$ for $0 \leq s \leq t_1 - t_0$. If $u_0^\leftarrow \sim \rho_{t_1}$ and $(u_s^\leftarrow)_{0 \leq s \leq t_1 - t_0}$ solves the reverse-time probability-flow ODE (16), then $u_s^\leftarrow \sim \rho_{\tau_s}$ for all $0 \leq s \leq t_1 - t_0$. In particular, $u_{t_1 - t_0}^\leftarrow \sim \rho_{t_0}$. If the flow is well defined down to $t_0 = 0$, then*

$$u_{t_1}^\leftarrow \sim \rho_0 = \rho_\vartheta(\cdot | \mathcal{C}), \quad T_\vartheta(u_{t_1}^\leftarrow) \sim \pi_\vartheta(\cdot | \mathcal{C}).$$

Proof. The marginals $(\rho_t)_{0 \leq t < 1}$ are generated by the forward VP-SDE when the initial law is $\rho_0 = \rho_\vartheta(\cdot | \mathcal{C})$. Indeed, for any bounded measurable test function φ ,

$$\begin{aligned} \mathbb{E}_{\xi_0 \sim \rho_0} [\varphi(\xi_t)] &= \frac{1}{Z} \mathbb{E}_{\xi_0 \sim r, \varepsilon \sim r} [\Lambda(\xi_0) \varphi(\alpha(t)\xi_0 + \sigma(t)\varepsilon)] \\ &= \frac{1}{Z} \mathbb{E}_{u \sim r} [\varphi(u) \mathbb{E}[\Lambda(\xi_0) | \xi_t = u]] = \int_{\mathbb{R}^{m_\xi}} \varphi(u) \rho_t(u) du. \end{aligned}$$

Thus ρ_t solves the Fokker-Planck equation

$$\partial_t \rho_t = -\nabla \cdot (b_t \rho_t) + \frac{1}{2} \beta(t) \Delta \rho_t, \quad b_t(u) = -\frac{1}{2} \beta(t) u. \quad (19)$$

Writing $s_t(u) = \nabla_u \log \rho_t(u)$, we can rewrite (19) as

$$\partial_t \rho_t = -\nabla \cdot (b_t \rho_t) + \frac{1}{2} \beta(t) \nabla \cdot (\rho_t s_t) = -\nabla \cdot \left(\left[b_t - \frac{1}{2} \beta(t) s_t \right] \rho_t \right).$$

It follows that ρ_t also solves the continuity equation

$$\partial_t \rho_t + \nabla \cdot (v_t \rho_t) = 0, \quad v_t(u) = b_t(u) - \frac{1}{2} \beta(t) s_t(u).$$

Meanwhile, substituting (18), we have that

$$v_t(u) = -\frac{1}{2} \beta(t) u - \frac{1}{2} \beta(t) (g(t, u) - u) = -\frac{1}{2} \beta(t) g(t, u).$$

This is exactly the vector field in the forward probability-flow ODE (15). By Assumption C.1, the ODE flow is well posed and gives the unique solution of the continuity equation. Therefore the backward flow (16) transports ρ_{t_1} to ρ_{t_0} . Taking $t_0 = 0$ gives $u_{t_1}^\leftarrow \sim \rho_0 = \rho_\vartheta(\cdot | \mathcal{C})$, and the process-space statement follows from Proposition 2.2. $\square$

Remark C.4. If $B(1) = \infty$, then $\alpha(t) \downarrow 0$ and the noised target marginals satisfy $\rho_t \rightarrow r$ in total variation as $t \uparrow 1$, provided $\Lambda \in L^1(r)$. In this sense, starting from $u_0^\leftarrow \sim r$ can be viewed as the limiting infinite-noising version of Theorem C.3. For a finite noising horizon, exact sampling from a terminal time t_1 requires initialising from the terminal marginal ρ_{t_1} , not from r . If one instead starts from r , the initialisation discrepancy is $\|\rho_{t_1} - r\|_{\text{TV}}$, and the discrepancy after exact reverse probability flow is at most this quantity, with equality when the deterministic flow is invertible.

D APPROXIMATION AND STABILITY GUARANTEES

The practical sampler replaces the exact guidance field $g(t, u)$ by an approximate field $\hat{g}_S(t, u)$, and then solves the resulting ODE or SDE numerically. This introduces two sources of error: the guidance approximation error and the numerical discretisation error. Both errors enter the probability-flow ODE and the reverse-time SDE through the same perturbation mechanism. We thus state the main stability and numerical-error bounds in a unified form.

D.1 COMMON PERTURBATION FRAMEWORK

We work on a truncated forward-time interval $[\tau, t_1]$, where $0 < \tau < t_1 < 1$, and use reverse-time notation throughout this subsection. Set

$$R_{\tau, t_1} = t_1 - \tau, \quad \kappa_s = t_1 - s, \quad 0 \leq s \leq R_{\tau, t_1}.$$

Thus s denotes reverse time and κ_s denotes the corresponding forward noising time. We index the two reverse dynamics by $\chi \in \{\text{ode}, \text{sde}\}$, and define

χ	λ_χ	ρ_χ	$\Sigma_\chi(s)$
ode	$\frac{1}{2}$	0	0
sde	1	$\frac{1}{2}$	$\sqrt{\beta(\kappa_s)} I_{m_\xi}$.

For an exact guidance field g and an approximate guidance field $\hat{g}$, define the exact and approximate reverse-time drifts

$$F_\chi(s, u) = \lambda_\chi \beta(\kappa_s) g(\kappa_s, u) - \rho_\chi \beta(\kappa_s) u, \quad \hat{F}_\chi(s, u) = \lambda_\chi \beta(\kappa_s) \hat{g}(\kappa_s, u) - \rho_\chi \beta(\kappa_s) u. \quad (20)$$

In particular, for the probability-flow ODE we have

$$\begin{aligned} a(s, u) &= F_{\text{ode}}(s, u) = \frac{1}{2} \beta(\kappa_s) g(\kappa_s, u), \\ \hat{a}(s, u) &= \hat{F}_{\text{ode}}(s, u) = \frac{1}{2} \beta(\kappa_s) \hat{g}(\kappa_s, u). \end{aligned}$$

Meanwhile, for the reverse-time SDE we have

$$\begin{aligned} b(s, u) &= F_{\text{sde}}(s, u) = \beta(\kappa_s) g(\kappa_s, u) - \frac{1}{2} \beta(\kappa_s) u, \\ \hat{b}(s, u) &= \hat{F}_{\text{sde}}(s, u) = \beta(\kappa_s) \hat{g}(\kappa_s, u) - \frac{1}{2} \beta(\kappa_s) u. \end{aligned}$$

The exact and approximate dynamics may then be written as

$$dX_s^\chi = F_\chi(s, X_s^\chi) ds + \Sigma_\chi(s) dW_s, \quad d\hat{X}_s^\chi = \hat{F}_\chi(s, \hat{X}_s^\chi) ds + \Sigma_\chi(s) dW_s. \quad (21)$$

When $\chi = \text{ode}$, we write $X^{\text{ode}} = u^\leftarrow$, $\hat{X}^{\text{ode}} = \hat{u}^\leftarrow$. When $\chi = \text{sde}$, we write $X^{\text{sde}} = \xi^\leftarrow$, $\hat{X}^{\text{sde}} = \hat{\xi}^\leftarrow$.

Let $\mathcal{R}_{\tau, t_1} \subset [\tau, t_1] \times \mathbb{R}^{m_\xi}$ be a region on which the exact and approximate guidance fields are compared, and which contains the exact and approximate trajectories under consideration. Define the realised pointwise guidance error by

$$\varepsilon(s) = \sup \left\{ \|g(\kappa_s, u) - \hat{g}(\kappa_s, u)\| : (\kappa_s, u) \in \mathcal{R}_{\tau, t_1} \right\}, \quad \delta_\chi(s) = \lambda_\chi \beta(\kappa_s) \varepsilon(s). \quad (22)$$

For the ODE and the SDE, we thus have that

$$\delta_{\text{ode}}(s) = \frac{1}{2} \beta(\kappa_s) \varepsilon(s), \quad \delta_{\text{sde}}(s) = \beta(\kappa_s) \varepsilon(s).$$

In the common case where $\varepsilon_{\tau, t_1} := \sup_{(t, u) \in \mathcal{R}_{\tau, t_1}} \|g(t, u) - \hat{g}(t, u)\| < \infty$, it follows from (22) that $\delta_\chi(s) \leq \lambda_\chi \beta(\kappa_s) \varepsilon_{\tau, t_1}$.

Given a time-dependent one-sided Lipschitz rate $\eta \in L^1([0, R_{\tau, t_1}])$, we define the integral

$$\Theta_\eta(s, r) = \exp \left\{ \int_r^s \eta(q) dq \right\}, \quad 0 \leq r \leq s \leq R_{\tau, t_1}.$$

Finally, throughout the statements below, we treat $\hat{g}$ as a realised approximate guidance field. If $\hat{g}$ is random, our results are to be read conditionally on this realisation. Probabilistic rates follow by combining these conditional stability bounds with separate probabilistic control of the realised guidance error.

D.2 UNIFIED STABILITY OF APPROXIMATE GUIDANCE

The following result is a variable-rate version of the deterministic perturbation bound obtained in (Moss et al., 2026, Proposition E.5), restated in our present notation and extended to cover the reverse-time SDE under a synchronous coupling.

Proposition D.1. *Fix $\chi \in \{\text{ode}, \text{sde}\}$, and let X^χ and $\hat{X}^\chi$ solve (21) on $[0, R_{\tau, t_1}]$. For the SDE, assume that the two equations are driven by the same Brownian motion. Assume that $(\kappa_s, X_s^\chi), (\kappa_s, \hat{X}_s^\chi) \in \mathcal{R}_{\tau, t_1}$ for all $s \in [0, R_{\tau, t_1}]$. Suppose that the exact reverse-time drift satisfies the one-sided Lipschitz condition*

$$\langle x - y, F_\chi(s, x) - F_\chi(s, y) \rangle \leq \eta_\chi(s) \|x - y\|^2 \quad (23)$$

whenever (κ_s, x) and (κ_s, y) belong to $\mathcal{R}_{\tau, t_1}$. Then, for every $0 \leq s \leq R_{\tau, t_1}$, the following bound holds:

$$\|X_s^\chi - \hat{X}_s^\chi\| \leq \Theta_{\eta_\chi}(s, 0) \|X_0^\chi - \hat{X}_0^\chi\| + \int_0^s \Theta_{\eta_\chi}(s, r) \delta_\chi(r) dr. \quad (24)$$

If the exact and approximate solves are coupled by the same reverse-time initial point, equivalently by the same terminal point at forward time t_1 , the first term is absent. Moreover, for $p \geq 1$, provided the initial laws have finite p -th moments,

$$W_p(\mathcal{L}(X_s^\chi), \mathcal{L}(\hat{X}_s^\chi)) \leq \Theta_{\eta_\chi}(s, 0) W_p(\mathcal{L}(X_0^\chi), \mathcal{L}(\hat{X}_0^\chi)) + \int_0^s \Theta_{\eta_\chi}(s, r) \delta_\chi(r) dr.$$

Proof. Let $D_s = X_s^\chi - \hat{X}_s^\chi$. When $\chi = \text{ode}$, this is an ordinary deterministic difference equation. When $\chi = \text{sde}$, the Brownian terms cancel under the synchronous coupling. In either case, we have

$$D_s = D_0 + \int_0^s \{F_\chi(r, X_r^\chi) - \hat{F}_\chi(r, \hat{X}_r^\chi)\} dr.$$

We can write the integrand as $F_\chi(r, X_r^\chi) - \hat{F}_\chi(r, \hat{X}_r^\chi) = F_\chi(r, X_r^\chi) - F_\chi(r, \hat{X}_r^\chi) + F_\chi(r, \hat{X}_r^\chi) - \hat{F}_\chi(r, \hat{X}_r^\chi)$. The first term is controlled by (23). The second term satisfies

$$\|F_\chi(r, u) - \hat{F}_\chi(r, u)\| = \lambda_\chi \beta(\kappa_r) \|g(\kappa_r, u) - \hat{g}(\kappa_r, u)\| \leq \delta_\chi(r)$$

on the region under consideration. Hence the upper right-hand derivative satisfies

$$\frac{d^+}{ds} \|D_s\| \leq \eta_\chi(s) \|D_s\| + \delta_\chi(s).$$

The variable-coefficient Gronwall inequality gives (24). The Wasserstein estimate follows by coupling the initial variables optimally and, in the SDE case, using the same Brownian motion for the two SDEs. $\square$

Corollary D.2. *Suppose the conditions of Proposition D.1 hold. Suppose in addition that $\eta_\chi(s) \leq \bar{\eta}_{\chi, \tau, t_1}$ on $[0, R_{\tau, t_1}]$ and $\bar{\beta}_{\tau, t_1} = \text{ess sup}_{t \in [\tau, t_1]} \beta(t) < \infty$. Then, under common terminal coupling, it holds for all $s \in [0, R_{\tau, t_1}]$ that*

$$\|X_s^\chi - \hat{X}_s^\chi\| \leq \lambda_\chi \bar{\beta}_{\tau, t_1} \varepsilon_{\tau, t_1} \Psi_{\bar{\eta}_{\chi, \tau, t_1}}(s), \quad \Psi_\eta(r) = \begin{cases} \frac{e^{\eta r} - 1}{\eta}, & \eta \neq 0, \\ r, & \eta = 0. \end{cases}$$

Proof. Under common terminal coupling, the first term in (24) vanishes. Moreover, since $\eta_\chi(s) \leq \bar{\eta}_{\chi, \tau, t_1}$ and $\delta_\chi(s) \leq \lambda_\chi \bar{\beta}_{\tau, t_1} \varepsilon_{\tau, t_1}$, we have $\Theta_{\eta_\chi}(s, r) \leq \exp\{\bar{\eta}_{\chi, \tau, t_1}(s - r)\}$. Therefore

$$\|X_s^\chi - \hat{X}_s^\chi\| \leq \lambda_\chi \bar{\beta}_{\tau, t_1} \varepsilon_{\tau, t_1} \int_0^s \exp\{\bar{\eta}_{\chi, \tau, t_1}(s - r)\} dr = \lambda_\chi \bar{\beta}_{\tau, t_1} \varepsilon_{\tau, t_1} \Psi_{\bar{\eta}_{\chi, \tau, t_1}}(s).$$

$\square$

D.3 CURVATURE REGIMES

The sign of the one-sided Lipschitz rate η_χ determines the nature of the bound in [Proposition D.1](#) and [Corollary D.2](#). This, in turn, is determined by the curvature of the smoothed latent log-likelihood. This is the subject of the following proposition.

Proposition D.3. *Suppose that, uniformly for $t \in [\tau, t_1]$, g is one-sided Lipschitz with constant ℓ_{τ, t_1} on the truncated region:*

$$\langle x - y, g(t, x) - g(t, y) \rangle \leq \ell_{\tau, t_1} \|x - y\|^2 \quad (25)$$

Then [Proposition D.1](#) applies with

$$\eta_\chi(s) = \beta(\kappa_s) (\lambda_\chi \ell_{\tau, t_1} - \rho_\chi). \quad (26)$$

Suppose, more strongly, that uniformly for $t \in [\tau, t_1]$, g is strongly dissipative with constant $\mu_{\tau, t_1} \geq 0$, on the truncated region:

$$\langle x - y, g(t, x) - g(t, y) \rangle \leq -\mu_{\tau, t_1} \|x - y\|^2. \quad (27)$$

Then [Proposition D.1](#) applies with

$$\eta_\chi(s) = -(\rho_\chi + \lambda_\chi \mu_{\tau, t_1}) \beta(\kappa_s). \quad (28)$$

Proof. Recalling the definition of the reverse-time drift F_χ from (20), it is straightforward to compute

$$\langle x - y, F_\chi(s, x) - F_\chi(s, y) \rangle = \lambda_\chi \beta(\kappa_s) \langle x - y, g(\kappa_s, x) - g(\kappa_s, y) \rangle - \rho_\chi \beta(\kappa_s) \|x - y\|^2.$$

The rates given in (26) and (28) now follow immediately after substituting (25) and (27), respectively. $\square$

D.4 GUIDANCE AND DISCRETISATION ERROR

The previous stability result controls the error between the exact guided dynamics and the exact approximately guided dynamics. It is also necessary to control the discretisation error introduced by a numerical solver.

Corollary D.4. *Assume that the hypotheses of [Proposition D.1](#) hold for the exact and approximately guided ODE and SDE dynamics. Let $0 = s_0 < \dots < s_N = R_{\tau, t_1}$ be a reverse-time grid with maximal step size h . Define*

$$A_{\eta_\chi} = \sup_{0 \leq s \leq R_{\tau, t_1}} \Theta_{\eta_\chi}(s, 0), \quad H_{\eta_\chi, \delta_\chi} = \sup_{0 \leq s \leq R_{\tau, t_1}} \int_0^s \Theta_{\eta_\chi}(s, r) \delta_\chi(r) dr.$$

Suppose that the numerical approximations $\hat{u}^{\leftarrow, h}$ and $\hat{\xi}^{\leftarrow, h}$ to the approximately guided ODE and SDE satisfy

$$\max_{0 \leq n \leq N} \|\hat{u}_{s_n}^{\leftarrow} - \hat{u}_{s_n}^{\leftarrow, h}\| \leq C_{\text{num}, \tau, t_1}^{\text{ode}} h^q, \quad \left\| \max_{0 \leq n \leq N} \|\hat{\xi}_{s_n}^{\leftarrow} - \hat{\xi}_n^{\leftarrow, h}\| \right\|_{L^p} \leq C_{\text{num}, p}^{\text{sde}} h^e. \quad (29)$$

Then it holds that

$$\begin{aligned} \max_{0 \leq n \leq N} \|u_{s_n}^{\leftarrow} - \hat{u}_{s_n}^{\leftarrow, h}\| &\leq A_{\eta_{\text{ode}}} \|u_0^{\leftarrow} - \hat{u}_0^{\leftarrow}\| + H_{\eta_{\text{ode}}, \delta_{\text{ode}}} + C_{\text{num}, \tau, t_1}^{\text{ode}} h^q, \\ \left\| \max_{0 \leq n \leq N} \|\xi_{s_n}^{\leftarrow} - \hat{\xi}_n^{\leftarrow, h}\| \right\|_{L^p} &\leq A_{\eta_{\text{sde}}} \|\xi_0^{\leftarrow} - \hat{\xi}_0^{\leftarrow}\|_{L^p} + H_{\eta_{\text{sde}}, \delta_{\text{sde}}} + C_{\text{num}, p}^{\text{sde}} h^e. \end{aligned} \quad (30)$$

In both displays, the initial-discrepancy term vanishes under common terminal coupling.

Proof. Apply [Proposition D.1](#) at the grid points and take the maximum over n . For the ODE, add the deterministic numerical error by the triangle inequality. For the SDE, first use the pathwise stability bound, then take the L^p -norm and apply Minkowski's inequality together with (29). $\square$

Corollary D.5. Assume that the hypotheses of [Corollary D.4](#) hold under a common terminal coupling, and consider a joint limit in which $S \rightarrow \infty$, $h \rightarrow 0$, and $a_S \rightarrow 0$. Suppose the realised guidance error satisfies $\varepsilon_{\tau, t_1} = O_{\mathbb{P}}(a_S)$ on $\mathcal{R}_{\tau, t_1}$. For $\chi \in \{\text{ode}, \text{sde}\}$, define

$$B_{\chi, \tau, t_1} = \sup_{0 \leq s \leq R_{\tau, t_1}} \int_0^s \Theta_{\eta_\chi}(s, r) \lambda_\chi(\kappa_r) dr.$$

Assume that the stability and numerical-error constants are uniformly controlled along this limit, in the sense that $B_{\chi, \tau, t_1} = O_{\mathbb{P}}(1)$, $C_{\text{num}, \tau, t_1}^{\text{ode}} = O_{\mathbb{P}}(1)$, $C_{\text{num}, p}^{\text{sde}} = O_{\mathbb{P}}(1)$. This is automatic when these quantities are deterministic and bounded independently of S and h . Then

$$\max_{0 \leq n \leq N} \|u_{s_n}^{\leftarrow} - \hat{u}_{s_n}^{\leftarrow, h}\| = O_{\mathbb{P}}(a_S + h^q), \quad \max_{0 \leq n \leq N} \|\xi_{s_n}^{\leftarrow} - \hat{\xi}_n^{\leftarrow, h}\| = O_{\mathbb{P}}(a_S + h^e).$$

In particular, if a uniform Monte Carlo guidance bound gives $a_S = S^{-1/2}$, then the corresponding rates are $O_{\mathbb{P}}(S^{-1/2} + h^q)$ for the ODE and $O_{\mathbb{P}}(S^{-1/2} + h^e)$ for the SDE.

Proof. Under a common terminal coupling, the initial-discrepancy terms in [Corollary D.4](#) vanish. Moreover,

$$H_{\eta_\chi, \delta_\chi} \leq B_{\chi, \tau, t_1} \varepsilon_{\tau, t_1} = O_{\mathbb{P}}(a_S).$$

For the ODE, the numerical term satisfies $C_{\text{num}, \tau, t_1}^{\text{ode}} h^q = O_{\mathbb{P}}(h^q)$, so the first conclusion follows from [\(30\)](#). For the SDE, the conditional L^p estimate in [\(29\)](#), together with $C_{\text{num}, p}^{\text{sde}} = O_{\mathbb{P}}(1)$, implies that

$$\max_{0 \leq n \leq N} \|\hat{\xi}_{s_n}^{\leftarrow} - \hat{\xi}_n^{\leftarrow, h}\| = O_{\mathbb{P}}(h^e).$$

due to Markov's inequality. The second conclusion follows from the triangle inequality. $\square$

D.5 PROCESS-SPACE PROPAGATION

The latent bounds pass directly to process space whenever the generator is Lipschitz on the region visited by the coupled latent trajectories.

Corollary D.6. Assume the hypotheses of [Corollary D.4](#). Suppose that E_m is equipped with a norm and that $T : \mathbb{R}^{m_\varepsilon} \rightarrow E_m$ is L_T -Lipschitz on the latent region visited by the exact, approximately guided, and numerical trajectories. Then the ODE and SDE bounds in [Corollary D.4](#) remain valid after applying T , with the right-hand sides multiplied by L_T . In particular,

$$\begin{aligned} \max_{0 \leq n \leq N} \|T(u_{s_n}^{\leftarrow}) - T(\hat{u}_{s_n}^{\leftarrow, h})\| &\leq L_T (A_{\eta_{\text{ode}}} \|u_0^{\leftarrow} - \hat{u}_0^{\leftarrow}\| + H_{\eta_{\text{ode}}, \delta_{\text{ode}}} + C_{\text{num}, \tau, t_1}^{\text{ode}} h^q), \\ \left\| \max_{0 \leq n \leq N} \|T(\xi_{s_n}^{\leftarrow}) - T(\hat{\xi}_n^{\leftarrow, h})\| \right\|_{L^p} &\leq L_T (A_{\eta_{\text{sde}}} \|\xi_0^{\leftarrow} - \hat{\xi}_0^{\leftarrow}\|_{L^p} + H_{\eta_{\text{sde}}, \delta_{\text{sde}}} + C_{\text{num}, p}^{\text{sde}} h^e). \end{aligned}$$

Moreover, any Wasserstein bound above is inherited under pushforward, since $W_p(T_{\#}\mu, T_{\#}\nu) \leq L_T W_p(\mu, \nu)$ whenever the relevant latent coupling is supported on the Lipschitz region.

Proof. The pathwise and strong L^p statements follow from $\|T(x) - T(y)\| \leq L_T \|x - y\|$. The Wasserstein statement follows by pushing forward the same latent coupling through T . $\square$

D.6 ADDITIONAL PATH-LAW CONTROL FOR THE REVERSE SDE

The preceding bounds control coupled paths. For the reverse-time SDE, one also obtains a path-law bound by Girsanov's theorem.

Proposition D.7. Assume that the usual Girsanov conditions hold, and let $\mathbb{P}$ and $\hat{\mathbb{P}}$ denote the path laws of the exact and approximately guided reverse SDEs on $C([0, R_{\tau, t_1}]; \mathbb{R}^{m_\varepsilon})$. Assume also that $\mathcal{L}(\xi_0^{\leftarrow}) \ll \mathcal{L}(\hat{\xi}_0^{\leftarrow})$. Then

$$\text{KL}(\mathbb{P} \parallel \hat{\mathbb{P}}) = \text{KL}(\mathcal{L}(\xi_0^{\leftarrow}) \parallel \mathcal{L}(\hat{\xi}_0^{\leftarrow})) + \frac{1}{2} \mathbb{E}_{\mathbb{P}} \int_0^{R_{\tau, t_1}} \beta(\kappa_s) \|g(\kappa_s, \xi_s^{\leftarrow}) - \hat{g}(\kappa_s, \xi_s^{\leftarrow})\|^2 ds. \quad (31)$$

In particular, if the two SDEs start from the same initial law, then the first term vanishes. If the realised guidance error is uniformly bounded by ε_{τ, t_1} along the exact paths, then, for $0 \leq s \leq R_{\tau, t_1}$,

$$\text{KL}(\mathbf{P}_{[0, s]} \| \widehat{\mathbf{P}}_{[0, s]}) \leq \text{KL}(\mathcal{L}(\xi_0^{\leftarrow}) \| \mathcal{L}(\widehat{\xi}_0^{\leftarrow})) + \frac{1}{2} \varepsilon_{\tau, t_1}^2 \int_0^s \beta(\kappa_r) \, dr.$$

Consequently,

$$\|\mathcal{L}(\xi_s^{\leftarrow}) - \mathcal{L}(\widehat{\xi}_s^{\leftarrow})\|_{\text{TV}}^2 \leq \frac{1}{2} \text{KL}(\mathcal{L}(\xi_0^{\leftarrow}) \| \mathcal{L}(\widehat{\xi}_0^{\leftarrow})) + \frac{1}{4} \varepsilon_{\tau, t_1}^2 \int_0^s \beta(\kappa_r) \, dr.$$

Proof. The two SDEs have a common diffusion coefficient. Moreover, we have from the definitions that

$$\|\Sigma_{\text{sde}}(s)^{-1}(b(s, u) - \widehat{b}(s, u))\|^2 = \beta(\kappa_s) \|g(\kappa_s, u) - \widehat{g}(\kappa_s, u)\|^2.$$

The identity in (31) follows immediately from Girsanov's theorem. The KL bound follows from the uniform guidance-error assumption. The total-variation bound follows by data processing and Pinsker's inequality. $\square$

D.7 DERIVATIVE-FREE GUIDANCE

When $\nabla_{\xi} \log \Lambda(\xi)$ is unavailable, one can estimate the guidance field using a form of Fisher's identity. Under the tilted bridge conditional density, we have

$$p(\xi_0 \mid \xi_t = u, \mathcal{C}) \propto \Lambda(\xi_0) \mathcal{N}(\xi_0; \alpha(t)u, \sigma(t)^2 I_{m_{\xi}}).$$

Using Fisher's identity, it follows that

$$\nabla_u \log \rho_t(u) = -\frac{1}{\sigma(t)^2} (u - \alpha(t) \mathbb{E}[\xi_0 \mid \xi_t = u, \mathcal{C}]).$$

Meanwhile, since $\rho_t(u) \propto \psi(t, u)r(u)$, we have $\nabla_u \log \rho_t(u) = g(t, u) - u$. Combining this with the previous expression, we arrive at

$$g(t, u) = \frac{\alpha(t)}{\sigma(t)^2} (\mathbb{E}[\xi_0 \mid \xi_t = u, \mathcal{C}] - \alpha(t)u).$$

This immediately suggests a gradient-free estimator. For a given pair (t, u) , draw bridge samples

$$\xi^{(s)} = \alpha(t)u + \sigma(t)\epsilon^{(s)}, \quad \epsilon^{(s)} \sim \mathcal{N}(0, I_{m_{\xi}}), \quad s = 1, \dots, S.$$

We can then define the self-normalised approximation

$$\widehat{g}_S^{\text{df}}(t, u) = \frac{\alpha(t)}{\sigma(t)^2} \left(\sum_{s=1}^S \bar{w}_s \xi^{(s)} - \alpha(t)u \right), \quad \bar{w}_s = \frac{\Lambda(\xi^{(s)})}{\sum_{\ell=1}^S \Lambda(\xi^{(\ell)})}.$$

This estimator only requires likelihood evaluations, but can be less sample-efficient than the gradient-based estimator because it estimates a small conditional-mean difference. Moreover, the prefactor $\alpha(t)/\sigma(t)^2$ can amplify Monte Carlo noise at small noising times.

E ADDITIONAL DETAILS FOR GUIDANCE-AWARE PARAMETER ESTIMATION

In this appendix, we fix the discretisation level m and suppress this from the notation. We thus write Z_{ϑ} for $Z_{\vartheta, m}$, and $\mathcal{J}(\vartheta) = \log Z_{\vartheta}$.

E.1 A STATIC INTERPRETATION VIA THE GIBBS–DONSKER–VARADHAN FORMULA

The evidence for the condition has a useful interpretation via the Gibbs–Donsker–Varadhan formula (Donsker & Varadhan, 1975).

Proposition E.1. Assume that $\Lambda_{\vartheta, \mathcal{C}} \geq 0$ and that $0 < Z_{\vartheta} < \infty$. Then the log evidence admits the representation

$$\log Z_{\vartheta} = \sup_{q \ll r} \{ \mathbb{E}_{\xi_0 \sim q} [\log \Lambda_{\vartheta, \mathcal{C}}(\xi_0)] - \text{KL}(q \| r) \}.$$

The supremum is attained at $\rho_{\vartheta}(\xi_0 | \mathcal{C})$, whenever the displayed quantities are finite. Moreover, whenever $q \ll \rho_{\vartheta}(\cdot | \mathcal{C})$,

$$\mathbb{E}_{\xi_0 \sim q} [\log \Lambda_{\vartheta, \mathcal{C}}(\xi_0)] - \text{KL}(q \| r) = \log Z_{\vartheta} - \text{KL}(q \| \rho_{\vartheta}(\cdot | \mathcal{C})). \quad (32)$$

Proof. The proof is standard. By definition, $\frac{\rho_{\vartheta}(\xi_0 | \mathcal{C})}{r(\xi_0)} = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi_0)}{Z_{\vartheta}}$. It follows that, whenever $q \ll \rho_{\vartheta}(\cdot | \mathcal{C})$,

$$\begin{aligned} \text{KL}(q \| \rho_{\vartheta}(\cdot | \mathcal{C})) &= \int \log \left(\frac{q(\xi_0)}{\rho_{\vartheta}(\xi_0 | \mathcal{C})} \right) q(\xi_0) d\xi_0 \\ &= \text{KL}(q \| r) - \mathbb{E}_{\xi_0 \sim q} [\log \Lambda_{\vartheta, \mathcal{C}}(\xi_0)] + \log Z_{\vartheta}. \end{aligned}$$

Rearranging this identity gives (32). The variational formula follows from nonnegativity of KL divergence, with equality at $q = \rho_{\vartheta}(\cdot | \mathcal{C})$. $\square$

By evaluating the identity in Proposition E.1 at $q = \rho_{\vartheta}(\cdot | \mathcal{C})$, we obtain

$$\log Z_{\vartheta} = \mathbb{E}_{\xi_0 \sim \rho_{\vartheta}(\cdot | \mathcal{C})} [\log \Lambda_{\vartheta, \mathcal{C}}(\xi_0)] - \text{KL}(\rho_{\vartheta}(\cdot | \mathcal{C}) \| r)$$

Combining this with the identity in Proposition B.2, we also have that

$$\log Z_{\vartheta} = \mathbb{E}_{f_0 \sim \pi_{\vartheta}(\cdot | \mathcal{C})} [\log L_{\mathcal{C}}(f_0)] - \text{KL}(\pi_{\vartheta}(\cdot | \mathcal{C}) \| p_{\vartheta}). \quad (33)$$

This shows that the evidence represents a balance between satisfaction of $\mathcal{C}$ and the relative-entropy cost of deforming p_{ϑ} into $\pi_{\vartheta}(\cdot | \mathcal{C})$.

Corollary E.2. Suppose that the additional condition $\mathcal{C}$ is the event that the discretised process lies in a feasible set $A \subseteq E_m$, so that $L_{\mathcal{C}}(f_0) = \mathbf{1}_A(f_0)$. If $p_{\vartheta}(A) > 0$, then $Z_{\vartheta} = p_{\vartheta}(A)$ and $\pi_{\vartheta}(\cdot | \mathcal{C}) = p_{\vartheta}(\cdot | A)$. Thus, in particular,

$$-\log Z_{\vartheta} = \text{KL}(p_{\vartheta}(\cdot | A) \| p_{\vartheta}).$$

Thus, maximising the evidence is equivalent to minimising the KL deformation from the prior process law to the constrained process law.

Proof. Since $L_{\mathcal{C}} = \mathbf{1}_A$, $Z_{\vartheta} = \int_A p_{\vartheta}(f_0) df_0 = p_{\vartheta}(A)$, and the conditioned law is $p_{\vartheta}(\cdot | A)$. Moreover,

$$\frac{dp_{\vartheta}(\cdot | A)}{dp_{\vartheta}}(f_0) = \frac{\mathbf{1}_A(f_0)}{p_{\vartheta}(A)}.$$

It follows that

$$\text{KL}(p_{\vartheta}(\cdot | A) \| p_{\vartheta}) = \mathbb{E}_{p_{\vartheta}(\cdot | A)} \left[\log \frac{1}{p_{\vartheta}(A)} \right] = -\log p_{\vartheta}(A) = -\log Z_{\vartheta}.$$

$\square$

Corollary E.3. Suppose that the additional condition is encoded by a likelihood of the form $L_{\mathcal{C}}(f_0) = a \exp\{-\Phi_{\mathcal{C}}(f_0)\}$, where $\Phi_{\mathcal{C}}(f_0) \geq 0$ and $a > 0$ is independent of ϑ . Then, up to additive constants independent of ϑ ,

$$-\log Z_{\vartheta} \equiv \text{KL}(\pi_{\vartheta}(\cdot | \mathcal{C}) \| p_{\vartheta}) + \mathbb{E}_{f_0 \sim \pi_{\vartheta}(\cdot | \mathcal{C})} [\Phi_{\mathcal{C}}(f_0)]. \quad (34)$$

Thus evidence maximisation balances a small deformation from the ordinary process law against good satisfaction of the soft condition.

Proof. By (33), we have that

$$\log Z_{\vartheta} = \mathbb{E}_{f_0 \sim \pi_{\vartheta}(\cdot | \mathcal{C})} [\log L_{\mathcal{C}}(f_0)] - \text{KL}(\pi_{\vartheta}(\cdot | \mathcal{C}) \| p_{\vartheta}).$$

Since $\log L_{\mathcal{C}}(f_0) = \log a - \Phi_{\mathcal{C}}(f_0)$, we obtain

$$-\log Z_{\vartheta} = -\log a + \mathbb{E}_{f_0 \sim \pi_{\vartheta}(\cdot | \mathcal{C})} [\Phi_{\mathcal{C}}(f_0)] + \text{KL}(\pi_{\vartheta}(\cdot | \mathcal{C}) \| p_{\vartheta}).$$

The term $-\log a$ is independent of ϑ , giving (34). $\square$

E.2 A DYNAMIC INTERPRETATION VIA DOOB TRANSFORM AND GIRSANOV'S THEOREM

The static free-energy identity shows that maximising Z_ϑ balances condition satisfaction against the KL cost of deforming the ordinary process law. We can also obtain a dynamic interpretation of this deformation cost, using Doob transforms and Girsanov's theorem (Doob, 1957; Girsanov, 1960; Øksendal, 2003). The result below shows that, under a suitable finite-horizon reference diffusion, the same deformation cost is represented by the energy of the guidance drift.

For each ϑ , recall that the latent likelihood is $\Lambda_{\vartheta, \mathcal{C}}$, the normalising constant is Z_ϑ , the reference density is $r(\xi) = \mathcal{N}(\xi; 0, I_{m_\xi})$, and the latent target is

$$\rho_\vartheta(\xi \mid \mathcal{C}) = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi)r(\xi)}{Z_\vartheta}.$$

We now give a path-space version of the same construction. Instead of tilting the static latent law r by $\Lambda_{\vartheta, \mathcal{C}}(\xi)$, we consider a stationary reference diffusion in latent space and tilt its path law by the likelihood of its terminal endpoint. This lets us interpret the KL deformation in terms of the energy of the Doob-transform guidance drift.

We work on a finite generation horizon, rescaled to $[0, 1]$. Let $\bar{\beta} \in L^1([0, 1])$ be nonnegative, and let $\mathbf{P}_\vartheta$ be the path law of the stationary reference latent Ornstein–Uhlenbeck process

$$dU_s = -\frac{1}{2}\bar{\beta}(s)U_s ds + \sqrt{\bar{\beta}(s)}dB_s, \quad U_0 \sim r, \quad 0 \leq s \leq 1. \quad (35)$$

Under $\mathbf{P}_\vartheta$, the one-time marginal of U_s is r for every $s \in [0, 1]$. Thus $\mathbf{P}_\vartheta$ is an unconditioned stationary reference dynamics in latent space. Define the tilted path law $\mathbf{Q}_\vartheta$ by

$$\frac{d\mathbf{Q}_\vartheta}{d\mathbf{P}_\vartheta} = \frac{\Lambda_{\vartheta, \mathcal{C}}(U_1)}{Z_\vartheta}.$$

Under $\mathbf{Q}_\vartheta$, paths are reweighted according to how well their terminal state satisfies the condition. The terminal marginal under $\mathbf{P}_\vartheta$ is r , while the terminal marginal under $\mathbf{Q}_\vartheta$ is precisely $\rho_\vartheta(\cdot \mid \mathcal{C})$. The corresponding Doob potential is

$$h_\vartheta(s, u) = \mathbb{E}_{\mathbf{P}_\vartheta} [\Lambda_{\vartheta, \mathcal{C}}(U_1) \mid U_s = u]. \quad (36)$$

This is the expected future condition likelihood from the current generation-time state u . Equivalently, h_ϑ is the same smoothed likelihood object as the noising-time potential ψ_ϑ , written in generation time rather than noising time.

Assumption E.4. For the fixed parameter value ϑ , assume that $0 < Z_\vartheta < \infty$, that $\Lambda_{\vartheta, \mathcal{C}} > 0$ r -almost everywhere, and that $\text{KL}(\rho_\vartheta(\cdot \mid \mathcal{C}) \parallel r) < \infty$. Assume that h_ϑ in (36) has a positive, sufficiently regular version for which $\nabla_u \log h_\vartheta$ is well defined, Itô's formula applies to $h_\vartheta(s, U_s)$, and the backward Kolmogorov drift term vanishes. Assume that the density process

$$M_{\vartheta, s} := \frac{h_\vartheta(s, U_s)}{Z_\vartheta}, \quad 0 \leq s \leq 1,$$

is a true $\mathbf{P}_\vartheta$ -martingale. Further, suppose that this process admits the stochastic exponential representation as $M_{\vartheta, s} = M_{\vartheta, 0} \mathcal{E}_{\vartheta, s}$, where $M_{\vartheta, 0} = \frac{h_\vartheta(0, U_0)}{Z_\vartheta}$ and

$$\mathcal{E}_{\vartheta, s} = \exp \left\{ \int_0^s \sqrt{\bar{\beta}(r)} \nabla_u^\top \log h_\vartheta(r, U_r) dB_r - \frac{1}{2} \int_0^s \bar{\beta}(r) \|\nabla_u \log h_\vartheta(r, U_r)\|^2 dr \right\}.$$

Theorem E.5. Under Assumption E.4, the tilted law $\mathbf{Q}_\vartheta$ is the Doob transform of $\mathbf{P}_\vartheta$ associated with h_ϑ . In particular, under $\mathbf{Q}_\vartheta$, the coordinate process U_s solves

$$dU_s = \left(-\frac{1}{2}\bar{\beta}(s)U_s + \bar{\beta}(s)\nabla_u \log h_\vartheta(s, U_s) \right) ds + \sqrt{\bar{\beta}(s)}dB_s^{\mathbf{Q}_\vartheta}, \quad (37)$$

where $B^{\mathbf{Q}_\vartheta}$ is Brownian motion under $\mathbf{Q}_\vartheta$, with initial law $U_0 \sim (\mathbf{Q}_\vartheta)_0$. Moreover,

$$\frac{1}{2} \mathbb{E}_{\mathbf{Q}_\vartheta} \left[\int_0^1 \bar{\beta}(s) \|\nabla_u \log h_\vartheta(s, U_s)\|^2 ds \right] = \text{KL}(\rho_\vartheta(\cdot \mid \mathcal{C}) \parallel r) - \text{KL}((\mathbf{Q}_\vartheta)_0 \parallel (\mathbf{P}_\vartheta)_0). \quad (38)$$

Proof. By the Markov property and Assumption E.4, h_ϑ solves the backward Kolmogorov equation associated with (35), with terminal condition $h_\vartheta(1, u) = \Lambda_{\vartheta, \mathcal{C}}(u)$. Using also Itô's formula, we have that

$$dh_\vartheta(s, U_s) = \sqrt{\bar{\beta}(s)} \nabla h_\vartheta(s, U_s)^\top dB_s.$$

The density process of $\mathbf{Q}_\vartheta$ with respect to $\mathbf{P}_\vartheta$ is

$$M_{\vartheta, s} = \mathbb{E}_{\mathbf{P}_\vartheta} \left[\frac{\Lambda_{\vartheta, \mathcal{C}}(U_1)}{Z_\vartheta} \mid \mathcal{F}_s \right] = \frac{h_\vartheta(s, U_s)}{Z_\vartheta}.$$

It follows that

$$\frac{dM_{\vartheta, s}}{M_{\vartheta, s}} = \sqrt{\bar{\beta}(s)} \nabla_u \log h_\vartheta(s, U_s)^\top dB_s.$$

By Girsanov's theorem,

$$B_s^{\mathbf{Q}_\vartheta} = B_s - \int_0^s \sqrt{\bar{\beta}(r)} \nabla_u \log h_\vartheta(r, U_r) dr$$

is Brownian motion under $\mathbf{Q}_\vartheta$. Substituting this into (35) gives (37). For the energy identity, write

$$\begin{aligned} \log M_{\vartheta, 1} &= \log M_{\vartheta, 0} + \int_0^1 \sqrt{\bar{\beta}(s)} \nabla_u \log h_\vartheta(s, U_s)^\top dB_s^{\mathbf{Q}_\vartheta} \\ &\quad + \frac{1}{2} \int_0^1 \bar{\beta}(s) \|\nabla_u \log h_\vartheta(s, U_s)\|^2 ds. \end{aligned}$$

Taking expectation under $\mathbf{Q}_\vartheta$, the stochastic integral has mean zero, and hence

$$\begin{aligned} \text{KL}(\mathbf{Q}_\vartheta \parallel \mathbf{P}_\vartheta) &= \mathbb{E}_{\mathbf{Q}_\vartheta} [\log M_{\vartheta, 1}] \\ &= \text{KL}((\mathbf{Q}_\vartheta)_0 \parallel (\mathbf{P}_\vartheta)_0) + \frac{1}{2} \mathbb{E}_{\mathbf{Q}_\vartheta} \int_0^1 \bar{\beta}(s) \|\nabla_u \log h_\vartheta(s, U_s)\|^2 ds. \end{aligned} \quad (39)$$

On the other hand, the likelihood ratio $d\mathbf{Q}_\vartheta/d\mathbf{P}_\vartheta$ depends only on U_1 . The left-hand side therefore simplifies to

$$\text{KL}(\mathbf{Q}_\vartheta \parallel \mathbf{P}_\vartheta) = \text{KL}((\mathbf{Q}_\vartheta)_1 \parallel (\mathbf{P}_\vartheta)_1).$$

Under $\mathbf{P}_\vartheta$, $U_1 \sim r$, and under $\mathbf{Q}_\vartheta$, $U_1 \sim \rho_\vartheta(\cdot \mid \mathcal{C})$. Substituting this into the previous display gives

$$\text{KL}(\mathbf{Q}_\vartheta \parallel \mathbf{P}_\vartheta) = \text{KL}(\rho_\vartheta(\cdot \mid \mathcal{C}) \parallel r).$$

Combining this with (39) proves (38). $\square$

Remark E.6. For a finite stationary OU horizon with finite integrated noise, U_0 and U_1 are generally correlated. Terminal reweighting can thus change the law of U_0 , resulting in the boundary term in (38). If $\int_0^1 \bar{\beta}(s) ds \rightarrow \infty$, then the correlation between U_0 and U_1 tends to zero. Equivalently, the initial generation-time state becomes independent of the endpoint. In that limit, terminal reweighting does not alter the initial law, and $(\mathbf{Q}_\vartheta)_0 = (\mathbf{P}_\vartheta)_0$.

We can now provide an interpretation of the evidence in terms of the guidance. Define the guidance energy by

$$\mathcal{E}_{\text{guide}}(\vartheta) = \frac{1}{2} \mathbb{E}_{\mathbf{Q}_\vartheta} \left[\int_0^1 \bar{\beta}(s) \|\nabla_u \log h_\vartheta(s, U_s)\|^2 ds \right].$$

Suppose that we are in the limiting regime where $(\mathbf{Q}_\vartheta)_0 = (\mathbf{P}_\vartheta)_0$. Then, by Theorem E.5, the guidance energy coincides with the relative-entropy cost

$$\mathcal{E}_{\text{guide}}(\vartheta) = \text{KL}(\rho_\vartheta(\cdot \mid \mathcal{C}) \parallel r).$$

It follows, combining this with the result in Proposition E.1, that the log evidence can be written as

$$\log Z_\vartheta = \mathbb{E}_{\xi_0 \sim \rho_\vartheta(\cdot \mid \mathcal{C})} [\log \Lambda_{\vartheta, \mathcal{C}}(\xi_0)] - \mathcal{E}_{\text{guide}}(\vartheta).$$

Thus evidence maximisation can be interpreted dynamically as balancing condition satisfaction against the guidance energy required to deform the reference latent process. For hard constraints, interpreted as limits of strictly positive soft constraints, the log-likelihood contribution vanishes on the constrained support, so evidence maximisation reduces, in the boundary-free regime, to minimising guidance energy. For soft constraints, the objective also includes the expected residual loss in Corollary E.3.

E.3 BASIC THEORETICAL GUARANTEES FOR PARAMETER ESTIMATION

We now record some basic theoretical results relating to our parameter estimation scheme. We begin by establishing that the exact fixed-parameter time slices give unbiased evidence gradients.

Proposition E.7. *Assume that $0 < Z_\vartheta < \infty$, that the reference density r and the noising schedule do not depend on ϑ , and that differentiation may be interchanged with integration. Then, for every $t \in [0, 1)$ such that $\psi_\vartheta(t, \cdot)$ is positive and differentiable in ϑ ,*

$$\nabla_\vartheta \mathcal{J}(\vartheta) = \nabla_\vartheta \log Z_\vartheta = \mathbb{E}_{u \sim \rho_{\vartheta, t}(\cdot | C)} [\nabla_\vartheta \log \psi_\vartheta(t, u)].$$

Thus, if a fixed-parameter LATENTFLOW solve at ϑ produces particles with selected time-slice marginals $u_j^{(i)} \sim \rho_{\vartheta, t_j}(\cdot | C)$, with $0 \leq t_j < 1$, then an unbiased estimator of $\nabla_\vartheta \mathcal{J}(\vartheta)$ is

$$\hat{G}(\vartheta) = \frac{1}{MJ} \sum_{i=1}^M \sum_{j=1}^J \nabla_\vartheta \log \psi_\vartheta(t_j, u_j^{(i)}).$$

Proof. Recall that $Z_\vartheta = \int_{\mathbb{R}^{m_\xi}} \psi_\vartheta(t, u) r(u) du$. Taking derivatives, and simplifying, we have

$$\begin{aligned} \nabla_\vartheta \log Z_\vartheta &= \frac{1}{Z_\vartheta} \int_{\mathbb{R}^{m_\xi}} \nabla_\vartheta \psi_\vartheta(t, u) r(u) du \\ &= \int_{\mathbb{R}^{m_\xi}} \nabla_\vartheta \log \psi_\vartheta(t, u) \frac{\psi_\vartheta(t, u) r(u)}{Z_\vartheta} du = \mathbb{E}_{u \sim \rho_{\vartheta, t}(\cdot | C)} [\nabla_\vartheta \log \psi_\vartheta(t, u)]. \end{aligned}$$

□

We now record an elementary convergence result for our update scheme. In practice, we cannot compute $h_\vartheta(t, u) := \nabla_\vartheta \log \psi_\vartheta(t, u)$ directly, and it must be replaced by a self-normalised ratio estimator $\hat{h}_{\vartheta, S}(t, u) := \nabla_\vartheta \log \psi_{\vartheta, S}(t, u)$; see [Section 3.3](#) for a related discussion. This estimator is generally biased for finite S , since it is a ratio of Monte Carlo averages. It is, however, consistent as $S \rightarrow \infty$, under the usual integrability assumptions and provided that the limiting denominator is positive. In any case, the following result only requires a conditional bias bound.

Theorem E.8. *Let $\Theta \subseteq \mathbb{R}^q$ be convex, and suppose that $\mathcal{J} : \Theta \rightarrow \mathbb{R}$ is continuously differentiable, bounded above by $\mathcal{J}^*$, and $L_{\mathcal{J}}$ -smooth along the iterates. Consider the updates*

$$\vartheta_{k+1} = \vartheta_k + \eta_k \hat{G}_k, \quad 0 < \eta_k < \frac{1}{2L_{\mathcal{J}}},$$

and assume that the iterates remain in Θ . Let $\mathcal{F}_k$ be the filtration generated by the algorithm before drawing $\hat{G}_k$. Suppose, in addition, that $\hat{G}_k$ admits the decomposition

$$\hat{G}_k = \nabla_\vartheta \mathcal{J}(\vartheta_k) + b_k + \xi_k,$$

where b_k is $\mathcal{F}_k$ -measurable, and satisfies $\|b_k\| \leq \beta_k$ for some $\mathcal{F}_k$ -measurable $\beta_k \geq 0$, and where ξ_k satisfies

$$\mathbb{E}[\xi_k | \mathcal{F}_k] = 0, \quad \mathbb{E}[\|\xi_k\|^2 | \mathcal{F}_k] \leq \sigma^2.$$

Then, for every $K \geq 1$,

$$\min_{0 \leq k < K} \mathbb{E}[\|\nabla_\vartheta \mathcal{J}(\vartheta_k)\|^2] \leq \frac{\mathcal{J}^* - \mathcal{J}(\vartheta_0) + \sum_{k=0}^{K-1} \eta_k \mathbb{E}[\beta_k^2] + \frac{L_{\mathcal{J}} \sigma^2}{2} \sum_{k=0}^{K-1} \eta_k^2}{\sum_{k=0}^{K-1} \frac{\eta_k}{2} (1 - 2L_{\mathcal{J}} \eta_k)}. \quad (40)$$

In particular, if $\eta_k \equiv \eta < 1/(2L_{\mathcal{J}})$ and $\beta_k \leq \beta$ for all k , the limiting stationarity floor is of order

$$O(\beta^2 + L_{\mathcal{J}} \sigma^2 \eta).$$

If, in addition, $\mathbb{E}[\beta_k^2] \rightarrow 0$, then the bias contribution to this floor vanishes; the remaining stochastic contribution can be made small by decreasing η , and disappears when $\sigma = 0$.

Proof. This follows from the standard smooth nonconvex analysis of stochastic gradient methods with biased gradients; see, for instance, [Ajallooeian & Stich \(2020\)](#). Applying the same descent-lemma argument to $f = -\mathcal{J}$, and using Young's inequality to absorb the bias term b_k , gives (40). □

F ADDITIONAL RELATED WORK

Conditioning stochastic-process priors. Perhaps the most well-studied case of stochastic-process conditioning is inference for Gaussian processes (GPs). Under finitely many linear-Gaussian observations, the posterior over any finite collection of function values remains Gaussian and can be computed analytically (Rasmussen & Williams, 2006). This conjugacy is generally lost under non-Gaussian likelihoods, motivating a large literature on approximate inference (Kuss & Rasmussen, 2005; Nickisch & Rasmussen, 2008; Hensman et al., 2015; Nickisch et al., 2018). Related work encodes known linear operator or differential-equation constraints directly in the GP prior or regression construction (Jidling et al., 2017; Lange-Hegermann, 2018; Gulian et al., 2022). By contrast, shape information such as monotonicity, boundedness, positivity, and convexity is commonly incorporated through virtual inequality observations, finite-dimensional coefficient inequalities, truncation, or posterior projection, generally resulting in non-Gaussian or projected posterior laws (Riihimäki & Vehtari, 2010; Da Veiga & Marrel, 2012; Lin & Dunson, 2014; Maatouk & Bay, 2017; López-Lopera et al., 2018; Agrell, 2019; Swiler et al., 2020; Maatouk et al., 2025; Astfalck et al., 2018). An alternative generalised Bayes linear approach enforces domain restrictions through the choice of inferential solution space (Astfalck et al., 2024). The closest GP-specific antecedent to the current work is FLOWGP (Moss et al., 2026), which conditions a finite-dimensional whitened GP representation on nonlinear or non-Gaussian information by solving a guided probability-flow ODE. Our framework contains FLOWGP as a special case but applies to a substantially broader class of stochastic-process priors; see Appendix A.

For SDE priors, LATENTFLOW is closely connected to diffusion-bridge simulation and conditioned diffusion processes. Classical bridge methods condition a diffusion to hit specified endpoints or endpoint observations, often by adding an approximate guiding term to its drift. Delyon & Hu (2006) construct readily simulated endpoint-guided diffusions and derive an explicit likelihood-ratio correction. Schauer et al. (2017) derive guiding terms from auxiliary tractable diffusions with known transition densities, while Bierkens et al. (2020) extend this guided-proposal framework to partial endpoint observations and elliptic or hypoelliptic settings. Related bridge samplers use coupling or other proposal mechanisms (Bladt & Sørensen, 2014; Bladt et al., 2016). More recent methods learn time reversals using score matching (Heng et al., 2025), estimate bridge scores without first learning the time reversal (Baker et al., 2025), develop infinite-dimensional conditioning constructions (Baker et al., 2024), introduce neural guided bridges (Yang et al., 2025b), or exploit Malliavin-calculus identities (Pidstrigach et al., 2025). Rare-event simulation and transition-path sampling address closely related path ensembles. Transition-path sampling uses path-space Monte Carlo to study rare reactive trajectories (Dellago et al., 1998; 2002), with recent diffusion-based methods also targeting transition-path ensembles (Seong et al., 2025), whereas genealogical particle methods and adaptive multilevel splitting estimate rare-event probabilities by repeatedly selecting and resampling trajectories according to their progress toward a rare set (Del Moral & Garnier, 2005; Cérou & Guyader, 2007; Cérou et al., 2012). LATENTFLOW targets likelihood-tilted path laws by tilting the law of the underlying innovations and transporting it toward regions of high condition likelihood. It therefore provides a likelihood-guided complement to output-space bridge proposals and splitting methods when suitable pulled-back gradients or guidance estimators are available.

Latent and reference-space formulations. Many stochastic-process laws admit representations as pushforwards of a tractable reference distribution rather than through an explicit density on the resulting process space. This perspective is implicit in simulator and filtering representations of state-space models (Doucet et al., 2001; Särkkä, 2013); in GMRF and SPDE constructions of spatial fields (Rue & Held, 2005; Lindgren et al., 2011); and in process priors and finite-dimensional representations built by composing or transforming latent random variables or functions, including random-feature kernel representations, latent-force models, GP state-space models, and deep GPs (Rahimi & Recht, 2007; Álvarez et al., 2009; 2013; Frigola et al., 2013; Damianou & Lawrence, 2013; Frigola et al., 2014). Pathwise GP constructions based on Matheron’s rule similarly represent posterior sample paths as transformations of prior samples and provide scalable samples for downstream Monte Carlo computation (Wilson et al., 2020; 2021). Innovation and non-centred parameterisations make

this explicit for diffusion processes by parameterising latent paths through transformed path variables or driving-noise variables together with model parameters (Roberts & Stramer, 2001; Golightly & Wilkinson, 2008; Papaspiliopoulos et al., 2013; Whitaker et al., 2017; Graham et al., 2022). Neural-process and implicit-process families likewise define learned distributions over functions through neural maps (Garnelo et al., 2018; Kim et al., 2019; Ma et al., 2019; Gordon et al., 2020). Similarly, normalising flows push forward simple latent laws, although they usually learn invertible transformations so that output densities remain available by change of variables (Rezende & Mohamed, 2015; Papamakarios et al., 2021).

The same reference-law viewpoint is central to Bayesian inverse problems. A posterior on a possibly infinite-dimensional parameter or function space is commonly defined by tilting a tractable reference measure by a likelihood (Stuart, 2010; Dashti & Stuart, 2017). Function-space MCMC methods, including pCN and Hilbert-space HMC, are designed directly for such targets and can remain well behaved under mesh refinement (Beskos et al., 2011; Cotter et al., 2013; Hairer et al., 2014). SMC, annealed importance sampling, and flow-augmented transport samplers construct sequences of intermediate distributions between reference and posterior laws (Neal, 2001; Del Moral et al., 2006; Arbel et al., 2021), while transport-map methods seek explicit reference-to-posterior maps, including in function-space inverse problems (El Moselhy & Marzouk, 2012; Hosseini et al., 2025). Recent score-based methods have likewise emphasised defining and approximating posterior sampling dynamics directly on function spaces (Baldassari et al., 2023; 2024; 2025).

Bayesian inference in generator noise space. A related literature performs posterior inference directly in the input-noise coordinates of a fixed generative model. For example, Holden et al. (2022) use parallel-tempered pCN in the latent space of a VAE, while Patel et al. (2022) use HMC together with a GAN. Another early contribution is Böhm et al. (2019). Normalising-flow priors have also been combined with variationally trained conditional flows for amortised inverse-problem inference (Whang et al., 2021). Related early work performs constrained-HMC inference in the input-noise space of a differentiable generator (Graham & Storkey, 2017), while distributional-control methods learn a transformation of the generator’s reference law toward specified output-space constraints (Wu et al., 2022).

There is also a substantial literature on combining transport maps with Monte Carlo methods. Transport-map methods construct invertible transformations between tractable reference laws and target laws, either as approximate posterior representations or as preconditioners and proposals for Monte Carlo sampling (El Moselhy & Marzouk, 2012; Parno & Marzouk, 2018; Hoffman et al., 2019; Cabezas & Nemeth, 2023; Cabezas et al., 2024). When an approximate map is embedded within a valid MCMC kernel with the appropriate correction, map approximation error affects sampling efficiency rather than the invariant target (Parno & Marzouk, 2018; Hoffman et al., 2019). NeuTra HMC uses a variationally trained normalising flow followed by HMC in the transported coordinates (Hoffman et al., 2019), whereas transport elliptical slice sampling uses a related nonlinear preconditioning strategy for elliptical slice sampling (Cabezas & Nemeth, 2023). A closely related latent construction is the neural-transport MCMC method of Nijkamp et al. (2022), in which an energy correction exponentially tilts the output law of a learned flow and sampling is performed in its latent coordinates. That work is directed toward learning and sampling energy-based generative models, rather than conditioning a stochastic-process simulator.

More recent methods apply the same principle to diffusion and flow-based generators. Outsourced diffusion sampling considers a deterministic representation $x = f_\vartheta(z)$, with z drawn from a Gaussian reference, and trains an amortised diffusion sampler for the latent posterior induced by an output-space likelihood or reward (Venkatraman et al., 2025). Noise-space HMC instead applies HMC directly to the initial noise of a deterministic reverse-diffusion map (Xia et al., 2026). Similar constructions have appeared for flow-matching models: Source-Guided Flow Matching modifies the source distribution while retaining the learned vector field, whereas D-Flow SGLD samples the source posterior induced by a new measurement operator (Wang et al., 2026; Parikh et al., 2026). Related inverse-problem methods perform SMC in the latent coordinates of diffusions, guide sampling trajectories under pre-trained latent flow priors, or combine latent diffusion posterior sampling with surrogate-likelihood guidance (Achituve et al., 2025; Askari et al., 2025; Wang & Tartakovsky, 2026).

These approaches share the central principle of conditioning a generator by changing the distribution of its input noise. LATENTFLOW differs primarily in the scope of the generator and in how the latent posterior is sampled. In our case, T_θ may describe an explicit stochastic-process prior, a numerical simulator, or a pre-trained neural generator, and need not be invertible or density-evaluable. Moreover, LATENTFLOW does not train an auxiliary conditional generator or amortised latent sampler. Instead, its time-dependent guidance is determined directly by the pulled-back likelihood and a known Gaussian noising bridge.

Diffusion guidance and finite-time controlled transport. The sampling construction underlying LATENTFLOW is closely related to diffusion and flow-based generative modelling. Diffusion probabilistic models define a forward noising process from a data distribution to a simple reference law together with a learned reverse process (Sohl-Dickstein et al., 2015; Ho et al., 2020); continuous-time score-based models generate samples by reversing a noising SDE or solving the associated probability-flow ODE (Song et al., 2021). Flow matching, rectified flows, and stochastic interpolants provide related continuous-time transport formulations (Lipman et al., 2023; Liu et al., 2023b; Albergo & Vanden-Eijnden, 2023; Albergo et al., 2025). Recent guided flow-matching methods extend these constructions to conditional generation and energy- or reward-guided sampling (Feng et al., 2025; Mark et al., 2025).

A large literature conditions pre-trained diffusion models on labels, measurements, inverse problems, or reward functions by modifying their reverse dynamics through classifier, classifier-free, measurement, reward, or off-the-shelf guidance terms (Dhariwal & Nichol, 2021; Ho & Salimans, 2022; Chung et al., 2023; Bansal et al., 2024). To control bias introduced by approximate guidance, other conditional diffusion samplers use SMC, particle MCMC, Feynman–Kac representations, twisting, or related correction mechanisms (Wu et al., 2023; Cardoso et al., 2024; Corenflos et al., 2025; Singhal et al., 2025; Skreta et al., 2025; Wu et al., 2025). Under their stated assumptions, some of these methods are asymptotically exact as the particle number increases, while others preserve a model-defined conditional target through a valid Monte Carlo correction; see Zhao et al. (2025) for a recent review. Diffusion and flow models have also been developed directly for function-valued data and function-space posterior sampling (Dutordoir et al., 2023; Franzese et al., 2023; Kerrigan et al., 2024; Lim et al., 2025; Yao et al., 2025).

These constructions admit a broader interpretation through finite-time stochastic control. Doob h -transforms provide a classical mechanism for conditioning Markov processes; for diffusions, the transformed dynamics acquire a drift correction involving the log-gradient of an appropriate space–time h -function (Doob, 1957; Didi et al., 2023). Related entropy-minimisation and Schrödinger-bridge formulations characterise controlled path measures as minimum-relative-entropy perturbations of a reference process subject to marginal or endpoint constraints (Föllmer, 1985; Léonard, 2014; Chen et al., 2016). Modern Schrödinger-bridge, Schrödinger–Föllmer, and controlled-diffusion samplers exploit this finite-time transport perspective for sampling unnormalised targets (Bernton et al., 2019; Huang et al., 2021; Zhang & Chen, 2022; Vargas et al., 2023b;a; 2024; Tamogashev & Malkin, 2026), while diffusion Schrödinger-bridge methods construct transports between prescribed endpoint laws (De Bortoli et al., 2021; Peluchetti, 2023; Liu et al., 2023a; 2024). Several recent conditional-diffusion methods make the connection to Doob transforms explicit (Didi et al., 2023; Denker et al., 2024; Guo et al., 2026). In infinite dimensions, Park et al. (2024) develop a function-space stochastic-control formulation of diffusion bridges; Yang et al. (2025a) learn discretisation-equivariant bridge scores using operator learning; and Pieper-Sethmacher et al. (2025) construct guided bridge measures for semilinear SPDEs conditioned on linear terminal observations. Baker et al. (2026) develop a complementary construction for conditioning trained function-space diffusion priors by supervised guidance.

LATENTFLOW instantiates the finite-time guidance principle in latent innovation space. Rather than learning a noising or transport process from samples, it uses a known reference-space bridge and the stochastic-process law supplied by T_θ . Its guiding field is the gradient of $\log h_t$, where h_t is the conditional expectation of the pulled-back likelihood under the latent noising bridge, equivalently the log-gradient of the corresponding smoothed pulled-back likelihood. Thus, LATENTFLOW combines the pullback formulation used in generator-noise inference with a diffusion-style finite-time transport, yielding a reference-space sampler for a broad class of stochastic-process priors.

G ADDITIONAL NUMERICAL EXPERIMENTS

G.1 SAMPLING BENCHMARKS

We now evaluate LATENTFLOW on three canonical benchmark distributions that are challenging for standard inference methods. Each target takes the form $p^*(x) \propto p_{\text{prior}}(x) \cdot p(y | x)$, with prior $p_{\text{prior}}(x) = \mathcal{N}(x; 0, I)$ and likelihoods $p(y | x)$ described below.

For LATENTFLOW, we use an Euler–Maruyama discretisation with 100 time steps, and 10 Monte Carlo samples for the guidance approximation. We benchmark against the No-U-Turn Sampler (NUTS; Hoffman & Gelman, 2014), mean-field VI (Blei et al., 2017), and a Laplace approximation. Sample quality is measured using the Maximum Mean Discrepancy (MMD²) with a multi-scale RBF kernel, with uncertainty reported as standard errors over 5 independent runs, alongside the effective sample size per second (ESS/s). We generate $N = 1,000$ samples using each method. NUTS thus requires 1,000 steps to produce its 1,000 samples, as well as an additional 500 warmup steps to ensure the chains are well mixed.

Across all benchmarks, LATENTFLOW achieves MMD² scores competitive with or better than NUTS, while producing *independent* samples. LATENTFLOW also succeeds on problems where standard MCMC collapses to a single mode.

Banana Distribution. The banana (twisted Gaussian) distribution has a curved, parabolic valley that resists random-walk exploration:

$$x_1 \sim \mathcal{N}(0, \sigma_1^2), \quad x_2 | x_1 \sim \mathcal{N}(b(x_1^2 - \sigma_1^2), \sigma_2^2),$$

with $b = 0.7$, $\sigma_1 = 1.0$, and $\sigma_2 = 0.4$. Figure 7 shows samples from each method overlaid on the true density. LATENTFLOW closely follows the curved valley, while VI and Laplace fail to capture the non-linear geometry. NUTS tracks the true distribution but produces far fewer independent samples, reflected in the low ESS of 67; see Table 2. LATENTFLOW achieves the best MMD², and is substantially faster than NUTS, achieving over 100 times greater ESS/s.

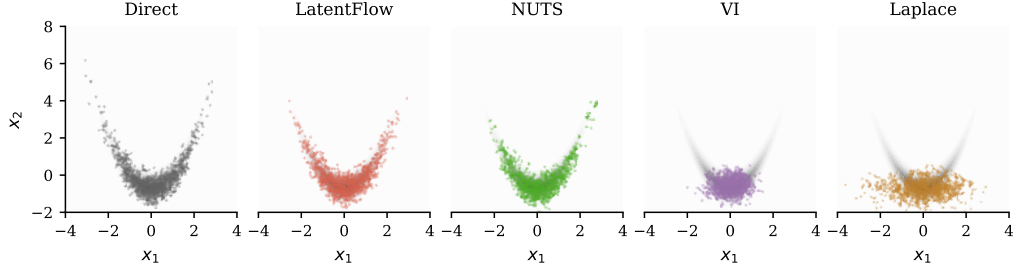


Figure 7: Banana distribution. Background: true density. Scatter: 1,000 samples from each method. Direct samples are exact draws. LATENTFLOW matches the true density well. VI and Laplace fail to capture the curved geometry.

Table 2: Banana distribution. MMD² $\pm$ stderr (lower is better), wall-clock time, and ESS/s, all over 5 independent runs (3 d.p.).

Method	MMD ²	Time (s)	ESS	ESS/s
LATENTFLOW	-0.001 ± 0.000	0.060 ± 0.000	1000	18010.0
NUTS	0.004 ± 0.003	0.470 ± 0.160	67	141.7
VI	0.059	0.310	1000	3270.7
Laplace	0.044	—	—	—

Ring Distribution. The ring (shell) distribution is concentrated on a thin annulus of radius R , with density given by

$$p^*(x) \propto \exp\left(-\frac{(\|x\|^2 - R^2)^2}{2\sigma^2}\right),$$

with $R = 2.0$ and $\sigma = 0.3$. Since gradients point predominantly *radially*, MCMC chains circulate slowly around the ring and mix poorly. As shown in Figure 8, LATENTFLOW traces the full annulus accurately and achieves over 20 $\times$ the ESS/s of NUTS.

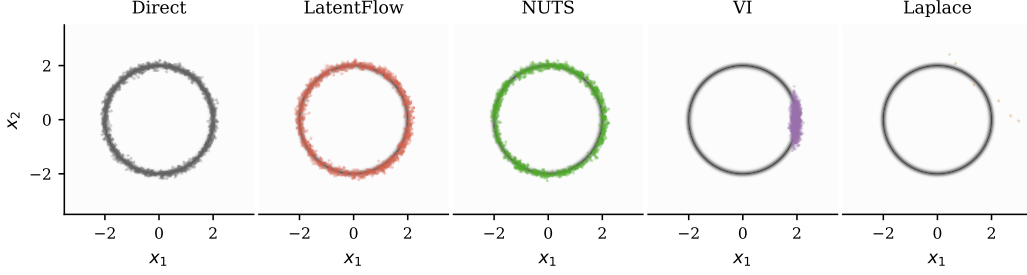


Figure 8: Ring distribution. Background: true density. Scatter: 1,000 samples from each method. LATENTFLOW traces the annulus accurately. NUTS also covers the ring but requires more warm-up steps. VI collapses to a central blob. Laplace diverges catastrophically.

Table 3: Ring distribution. $\text{MMD}^2 \pm \text{stderr}$ (lower is better), wall-clock time, ESS/s, and radial statistics, all over 5 independent runs (3 d.p.). Ground truth: $\mathbb{E}[r] = 2$, $\text{Std}[r] \approx 0.076$.

Method	MMD^2	Time (s)	ESS/s	$\mathbb{E}[r]$	$\text{Std}[r]$
LATENTFLOW	0.000 ± 0.000	0.050 ± 0.000	21263.4	2.004	0.086
NUTS	0.001 ± 0.001	0.340 ± 0.060	897.1	1.999	0.075
VI	0.453	0.290	3412.0	1.960	0.087
Laplace	0.458	—	—	274.105	209.312

Mixture of Gaussians. We test multi-modal coverage using a mixture of $K = 4$ equal-weight Gaussians centred at $(\pm 2, \pm 2)$ with $\sigma = 0.5$. LATENTFLOW places samples in all four quadrants, while NUTS concentrates in quadrant 4; see Figure 9. While NUTS achieves a raw ESS of 943 and an ESS/s of 2,191.4 (Table 4), its samples originate from a single mode. Its apparent efficiency is thus misleading: it fails to cover multiple modes, reflected in its MMD^2 of 0.401 ± 0.009 . LATENTFLOW achieves the best sample quality with $\text{MMD}^2 = 0.00079 \pm 0.00027$, recovering the correct global mean of approximately (0, 0) and standard deviation near 2.04 in both dimensions, compared to NUTS’s collapsed mean of (2.016, 1.984) with standard deviation ≈ 0.490 .

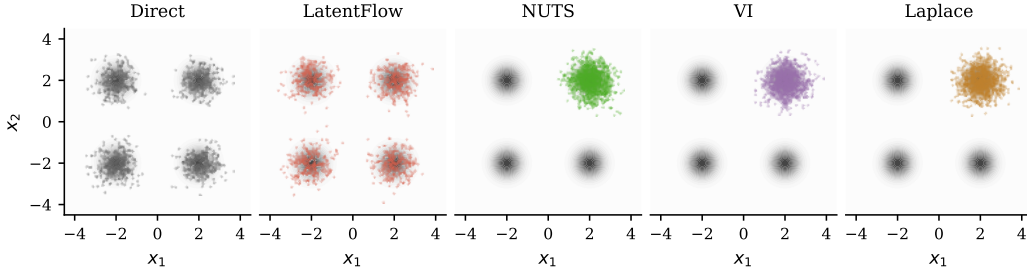


Figure 9: Mixture of Gaussians. Background: true density (four modes). Scatter: 1,000 samples from each method. LATENTFLOW covers all four modes. NUTS collapses entirely to quadrant 4 (100% of the chain). VI collapses to a single Gaussian centred near (2, 2).

Table 4: Mixture of Gaussians. $\text{MMD}^2 \pm \text{stderr}$ (lower is better), wall-clock time, and ESS/s, all over 5 independent runs (3 d.p.). Despite its high ESS/s, NUTS fails to achieve multi-modal coverage (all samples are from one mode).

Method	MMD^2	Time (s)	ESS	ESS/s
LATENTFLOW	0.001 ± 0.000	0.050 ± 0.000	1000	19865.9
NUTS	0.401 ± 0.009	0.430 ± 0.130	943	2191.4
VI	0.413	0.140	1000	7119.4

G.2 SPATIAL PROCESSES

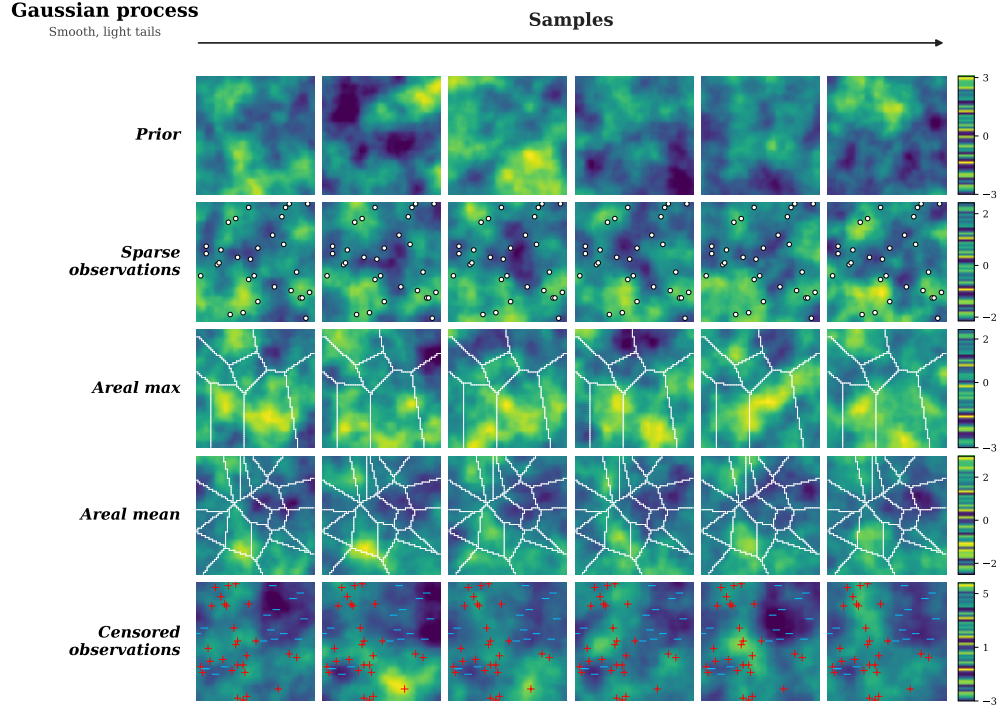


Figure 10: **Conditional sampling from a Gaussian process prior.** The base process is a smooth, light-tailed Gaussian field (Matérn-3/2 kernel, lengthscale 0.15). The posteriors interpolate smoothly between observations and honour each areal summary while preserving the short-lengthscale texture of the prior.

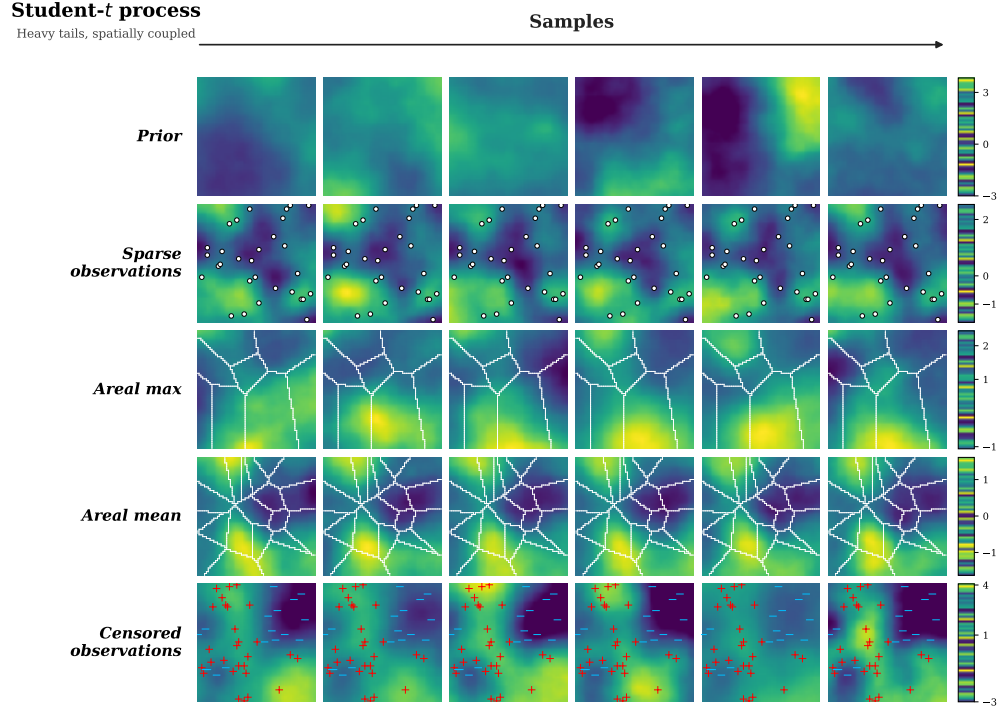


Figure 11: **Conditional sampling from a Student- t process prior.** The base process is a heavy-tailed, spatially coupled Student- t field ($\nu_t = 4$, Matérn-3/2 kernel, lengthscale 0.5).

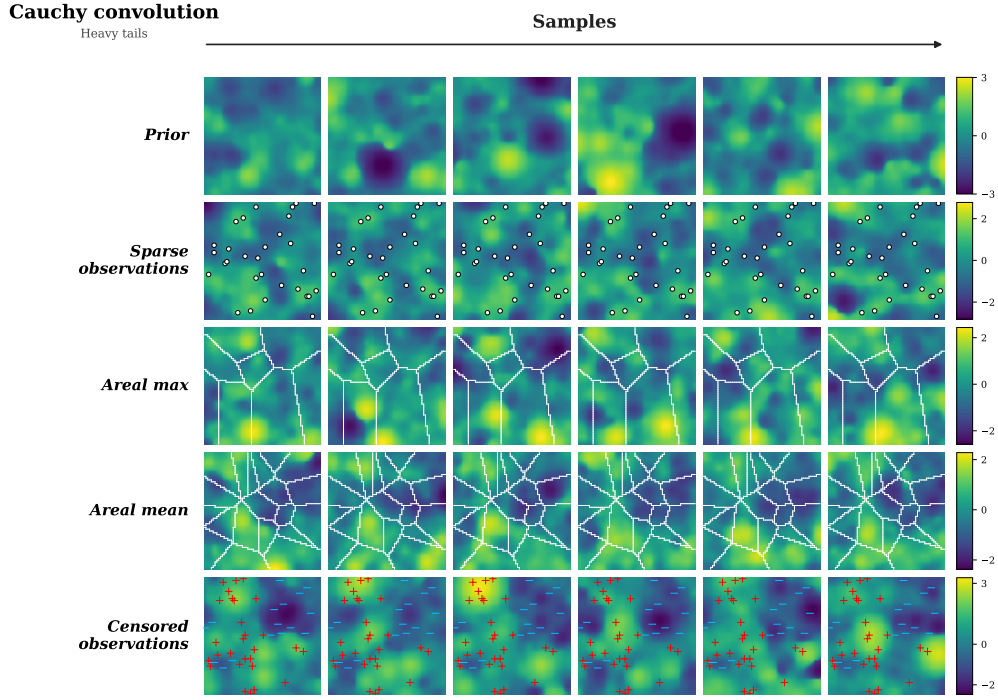


Figure 12: **Conditional sampling from a Cauchy convolution prior.** The base process is a very heavy-tailed field formed by convolving a Cauchy noise source with a short-lengthscale kernel (lengthscale 0.05). The infinite-variance source produces sharp, highly localised features; the posteriors reconcile these with the imposed point, areal, and censoring constraints.

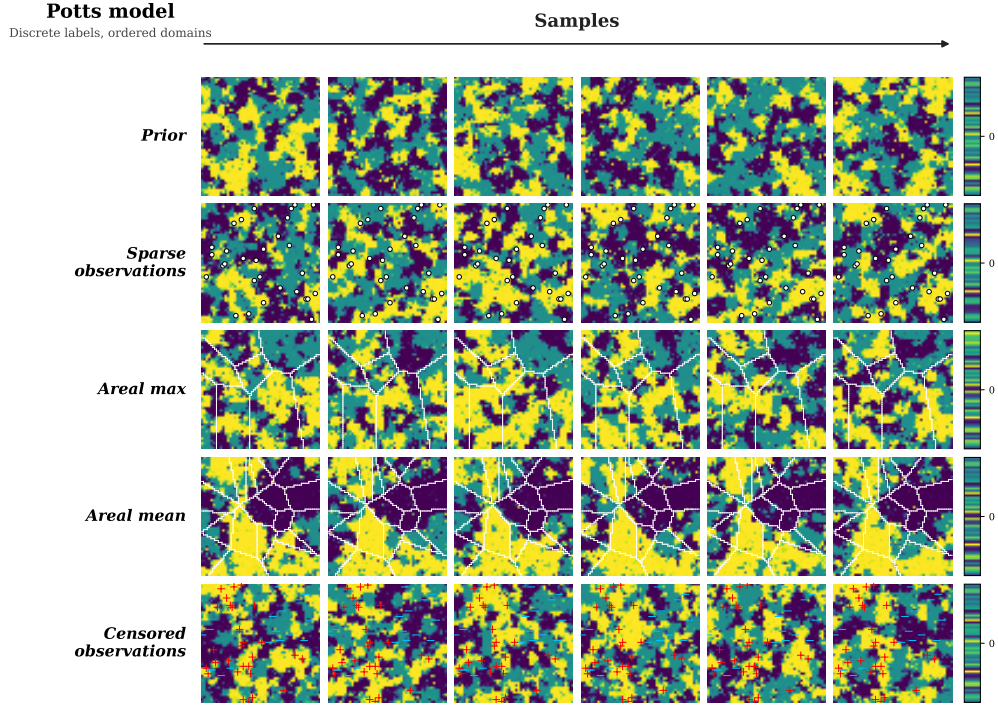


Figure 13: **Conditional sampling from a Potts model prior.** The base process is a discrete-label field with ordered domains (Potts model, $q = 3$ states mapped to $\{-1, 0, 1\}$, inverse temperature $\beta = 1.2$). Conditioning reshapes the domain structure to match the observations while retaining the piecewise-constant, label-coherent character of the prior.

G.3 TEMPORAL PROCESSES

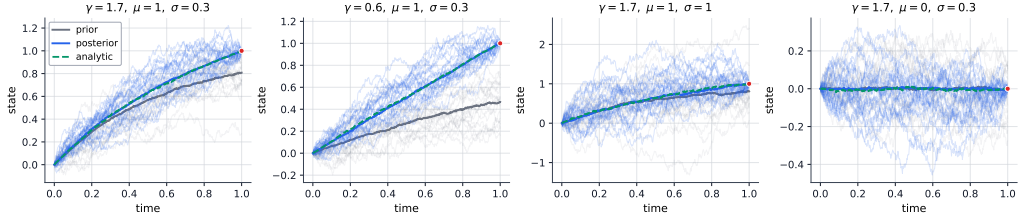


Figure 14: **Conditional sampling from an OU process prior.** The prior is an OU diffusion process with mean-reversion rate γ , long-run mean μ , and diffusion coefficient σ . Each panel shows prior trajectories, posterior bridge samples, and the analytic bridge mean for a different choice of (γ, μ, σ) . The red marker denotes the terminal constraint, blue curves denote LATENTFLOW posterior samples, grey curves denote prior samples, and the green dashed curve denotes the analytic bridge mean.

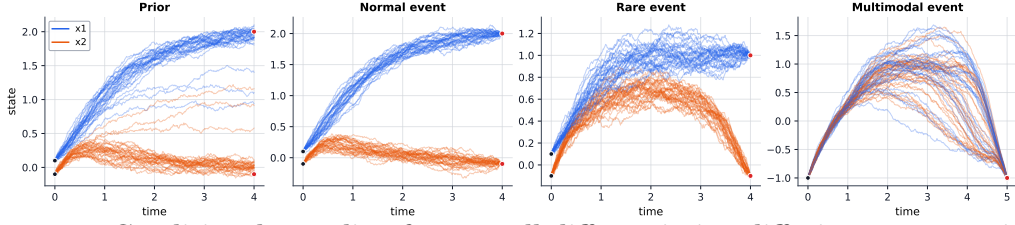


Figure 15: **Conditional sampling from a cell-differentiation diffusion process prior.** The panels show samples from the prior and from three endpoint-conditioned laws: a typical terminal event, a rare terminal event, and a multimodal terminal event. Blue and orange curves denote the two state variables x_1 and x_2 , black markers denote the initial state, and red markers denote the terminal constraint. The rare and multimodal examples illustrate that the same latent bridge construction can condition nonlinear diffusion priors on *very* low-probability events. In particular, for the rare event and the multimodal event no valid samples were found after simulating the unconditioned process 100,000 times.

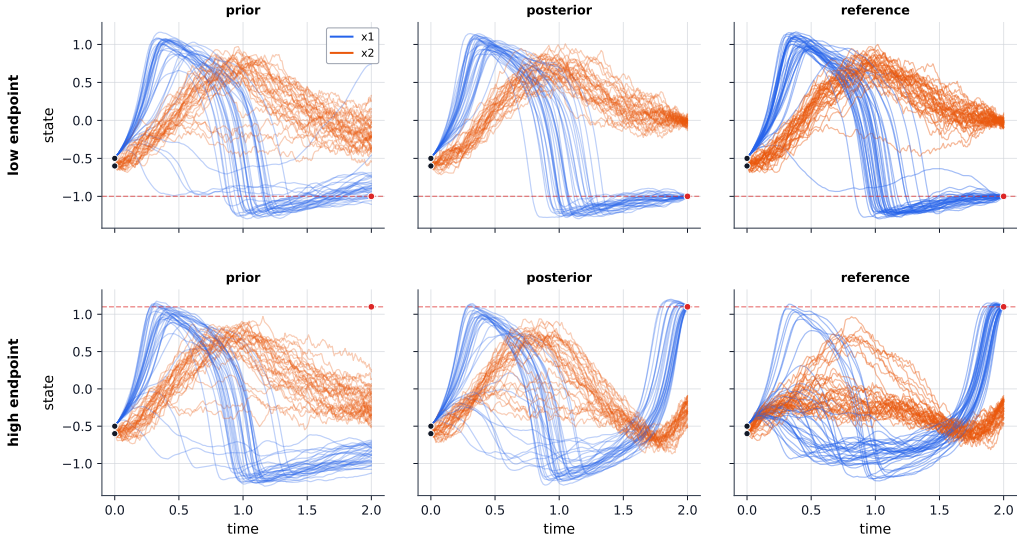


Figure 16: **Conditional sampling from a FitzHugh–Nagumo diffusion process prior.** Rows: two endpoint constraints on the first state variable: a low endpoint and a high endpoint. Columns: prior trajectories, LATENTFLOW posterior bridge samples, and reference bridge trajectories. Blue and orange curves denote the two state variables x_1 and x_2 , black markers denote the initial state, and red markers and dashed lines denote the terminal constraint. The high-endpoint case illustrates conditioning on a rare excursion of the excitable system.

G.4 SPATIO-TEMPORAL PROCESSES

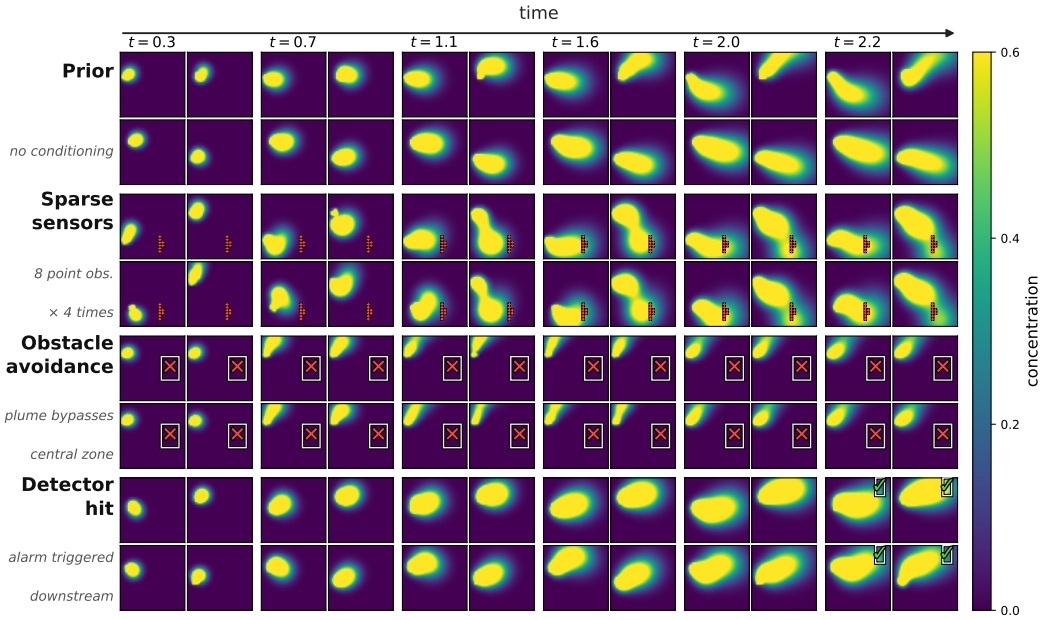


Figure 17: **Posterior samples from the advection–diffusion plume model under three conditioning regimes, each showing four independent draws at six time snapshots.** *Prior*: unconditioned samples exhibiting diverse plume trajectories and intensities. *Sparse sensors*: conditioning on exact concentration values at 8 spatial locations observed at 4 times (32 observations total) recovers the ground-truth plume trajectory. *Obstacle avoidance*: a binary condition requiring the plume to stay below concentration 0.06 in the central rectangular region (dashed box, $\times$) at all times steers samples around the obstacle. *Detector hit*: a binary condition requiring peak concentration to exceed 0.40 in the downstream detector region (dashed box, $\checkmark$) concentrates mass on trajectories that reach the top-right corner.

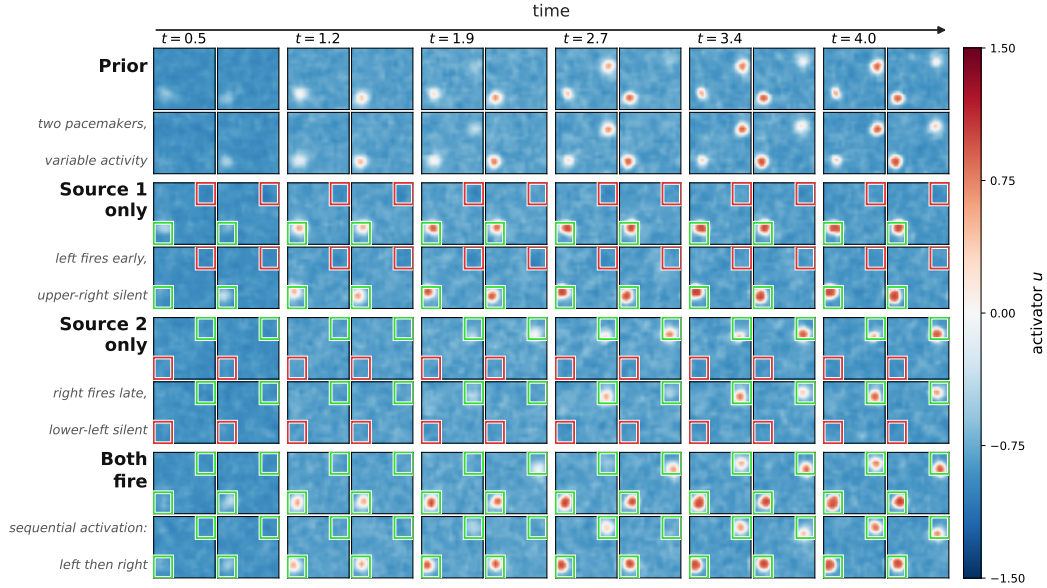


Figure 18: **Posterior samples from the FitzHugh–Nagumo excitable medium under three conditioning regimes, each showing four independent draws at six time snapshots.** *Prior*: unconditioned samples show variable activation of two spatially separated pacemakers (green boxes: active; red boxes: silent). *Source 1 only*: conditioning on the lower-left pacemaker firing early ($t \lesssim 1.3$) while the upper-right remains silent produces a single travelling wavefront originating from the lower-left corner. *Source 2 only*: the complementary condition suppresses the lower-left source and requires the upper-right to fire late ($t \gtrsim 1.8$), yielding a wavefront from the upper-right. *Both fire*: conditioning on sequential activation of both sources produces two distinct wavefronts propagating across the medium in succession.

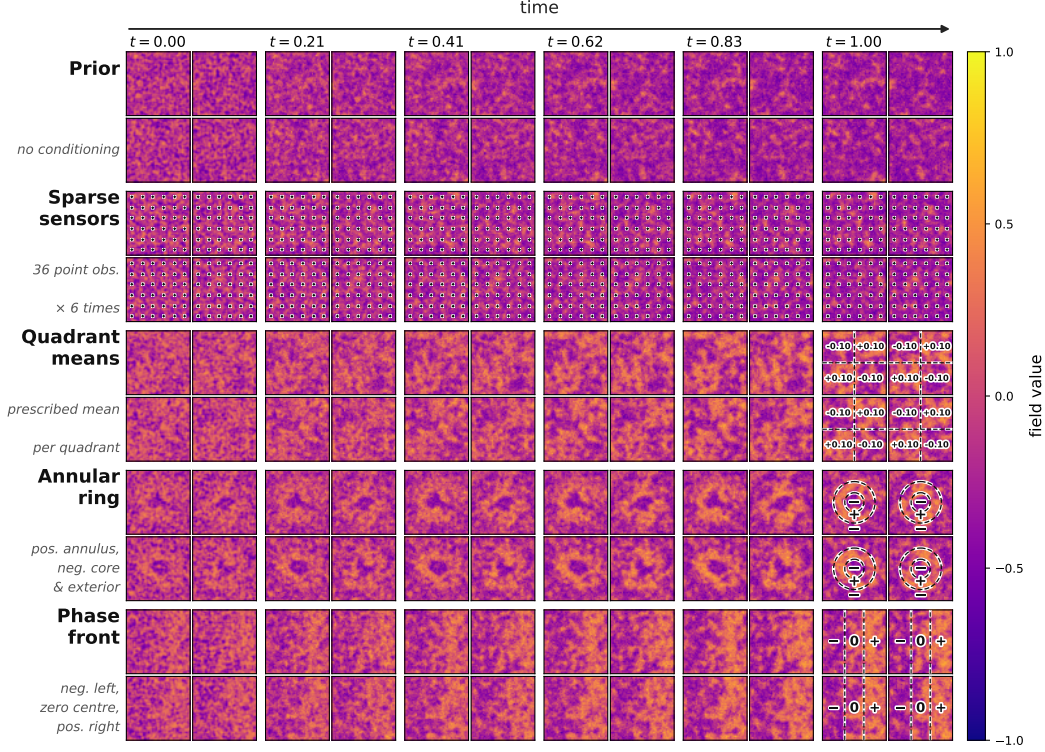


Figure 19: **Posterior samples from the Allen–Cahn phase-field model under four conditioning regimes, each showing four independent draws at six time snapshots.** Except for the sparse-sensor experiment, all conditions constrain statistics of the final field $u(\cdot, t_{29})$. *Prior*: unconditioned samples exhibit stochastic domain coarsening toward the ± 1 phases. *Sparse sensors*: conditioning on 36 point observations at each of 6 snapshot times (216 observations total) recovers the ground-truth field trajectory. *Quadrant means*: prescribing the spatial mean in each quadrant to a checkerboard pattern $(+0.10, -0.10, -0.10, +0.10)$ steers the final phase arrangement into the corresponding diagonal structure. *Annular ring*: prescribing a positive mean in an annular band $(0.30 \leq r \leq 0.65)$ and negative means in the core and exterior produces a ring-shaped positive domain surrounded by negative phase. *Phase front*: prescribing a left-to-right mean gradient $(-0.20, 0.0, +0.20)$ across three vertical strips $(|x| \leq 0.30)$ boundaries) induces a diffuse phase interface running vertically through the domain.